# CONVERGENCE OF GLOWINSKI'S DECOMPOSITION–COORDINATION METHOD FOR THE $g$-LAPLACIAN

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Abstract. We study Glowinski's decomposition–coordination method (augmented Lagrangian, ALG2/ADMM) for the Dirichlet problem for the $g$-Laplacian $-\operatorname{div}\big(g(x,|\nabla u|)\nabla u/|\nabla u|\big) = f$, where $g = \partial_t G$ and $G(x,\cdot)$ is an arbitrary convex Young function. Glowinski's classical theorem gives strong convergence when the energy splits off a uniformly convex differentiable part on $L^2$; we show that this covers $p \geq 2$ and uniformly convex $G$, but not $1 < p < 2$. Following Glowinski's Lyapunov argument, we prove weak convergence for every convex Young function, including non-$\Delta_2$ ones, under the assumption (S) that the augmented Lagrangian has a saddle point, and strong convergence in $W^{1,p}$ for $1 < p < 2$; without (S) the iterates are unbounded. Since (S) is a regularity property of the solution, we review when it is known to hold. We discuss the P1 problem and the Lavrentiev phenomenon, and report experiments with exact solutions.

## 1. Introduction

Let $\Omega \subset \mathbb{R}^d$ be a bounded open set and let $G\colon \Omega \times [0,\infty) \to [0,\infty)$ be such that $G(x,\cdot)$ is a convex Young function for a.e. $x$. We are interested in the numerical solution of

$$\min_v \; J(v), \qquad J(v) = \int_\Omega G(x,|\nabla v|)\,dx - \int_\Omega f v\,dx, \tag{1.1}$$

with homogeneous Dirichlet data, whose Euler–Lagrange equation is the *$g$-Laplace equation*

$$-\operatorname{div}\Big(g(x,|\nabla u|)\frac{\nabla u}{|\nabla u|}\Big) = f \quad \text{in } \Omega, \qquad u = 0 \quad \text{on } \partial\Omega, \qquad g = \partial_t G. \tag{1.2}$$

This class contains the $p$-Laplacian ($G = t^p/p$), the $p(x)$-Laplacian of electrorheological fluids [2], double phase and two-phase energies $t^p/p + a(x)t^q/q$, $a \geq 0$ [52, 22], Orlicz energies such as $t^p \log^a(1+t)$, and energies of exponential growth such as $e^t - 1 - t$, which violate the $\Delta_2$ condition.

The decomposition–coordination method of Glowinski and Marrocco [40] and Fortin and Glowinski [35, 36, 39] (algorithm ALG2; for the parameters used here, an alternating direction method of multipliers, ADMM [41, 14]) introduces the flux $\nu = \nabla u$ and alternates a *linear* Poisson problem for $u$, a *pointwise* nonlinear problem for $\nu$, and an update of the multiplier $\eta$. Since $G$ enters only through a scalar monotone equation in each element, the P1 code for the $p(x)$-Laplacian of [4] extends directly to arbitrary Young functions (package `glaplace`).

The classical convergence theory is Glowinski's [39, Ch. VI, Thm. 5.1] (following Fortin and Glowinski [36]): if the augmented Lagrangian has a saddle point, $F(q) = \int_\Omega G(x,|q|)\,dx$ is superlinear on $L^2$ and $F$ splits off a Gâteaux differentiable part that is uniformly convex on bounded sets, then $(u_n, \nabla u_n)$ converges strongly. On $H = L^2$ such a splitting exists for $t^p/p$ with $p \geq 2$ and, more generally, whenever $G(x,\cdot) - \frac{\varepsilon}{2}t^2$ is convex or $g(x,t)/t$ is nondecreasing, positive and bounded at $t = 1$ (Lemma 2.13); this justifies the application in [4, Thm. 4.3] for $p(x) \geq 2$. For $1 < p < 2$ the hypotheses fail (Proposition 2.14), and [39, Ch. VI, §6.3] applies the theorem to the $s$-Laplacian only after discretisation. We clarify what can be proved for $p < 2$ and non-polynomial growth, at the continuous and the discrete level.

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**Contributions.** *What is classical.* The algorithm; the Lyapunov inequality behind Theorem A (our identity (2.9) is [39, Ch. VI, (5.25)] with $\rho = r$, written as an equality); the weak convergence of the whole multiplier sequence via Opial's lemma [39, Ch. VI, §7]; the existence of discrete saddle points (Theorem C(ii); [39, Ch. VI, §4.2]) and finite-dimensional ADMM convergence (Theorem C(iii); [31, Thm. 8]). Theorem B(a) follows from Glowinski's theorem via Lemma 2.13 whenever $\operatorname{ess\,sup}_x g(x,1) < \infty$. We give complete proofs because the setting (arbitrary Young integrands, nonsmooth $F$, no growth conditions) and the exact hypotheses matter.

*What is new.* (1) The convergence theory under (S) alone, without Glowinski's hypotheses (iii)–(iv) of Section 2.6, for $\rho = r$: weak convergence for every convex Young integrand, including non-$\Delta_2/\nabla_2$ and nonsmooth ones (Theorem A); strong convergence in $W_0^{1,p}$ for $t^p/p$, $1 < p < 2$ (Theorem B(b)), and in $H_0^1$ for nondecreasing $g(x,t)/t$ also when $\operatorname{ess\,sup}_x g(x,1) = \infty$; unboundedness of the iterates when (S) fails (Proposition 2.11, an application of [31, Thm. 7]). (2) The scope of Glowinski's theorem on $L^2$ (Lemma 2.13, Proposition 2.14). (3) The explicit pointwise solver for arbitrary Young functions (Lemma 2.7). (4) The characterisation of (S) for the $p$-Laplacian (Proposition 2.16) and a review of when (S) is known to hold (Section 3, Table 1). (5) The consequences of the Lavrentiev phenomenon for the method, and, for our radial test problems, (S), a lower energy bound for conforming functions and convergence of the P1 minimisers (Remark 5.2, Lemma 5.1, Proposition 5.3). (6) Experiments with exact solutions and errors computed by quadrature (Section 6).

**What is not proved.** (i) (S) is a regularity property of the solution, available from the literature only in some cases (Section 3); when it fails, $(\eta_n, \nu_n)$ is unbounded (Proposition 2.11), but the behaviour of $u_n$ is open. (ii) We prove no rate, neither for the iteration nor for the finite element error; the orders in Section 6 are observations. (iii) We treat only $\rho = r$. (iv) We do not prove, in general, that the P1 minimisers $u_h$ converge to the continuous minimiser: Lemma 5.1 gives only the lower bound $J(v_h) \geq J(u)$ for conforming $v_h$, and Proposition 5.3 covers only the autonomous radial test problems (not the two-phase ones). (v) Proposition 2.16 and Section 4 are written for $d = 2$ (Proposition 2.16 holds in any dimension for $p \geq 2d/(d+2)$).

**Notation.** $(\cdot,\cdot)$ and $\|\cdot\|$ denote the inner product and norm of $L^2(\Omega)$ or $L^2(\Omega;\mathbb{R}^d)$; other norms carry a subscript. For a convex function $\Psi\colon X \to (-\infty, +\infty]$ on a Hilbert space, $\operatorname{dom}\Psi = \{\Psi < \infty\}$ and $\partial\Psi(q) = \{\zeta \in X : \Psi(q') \geq \Psi(q) + (\zeta, q' - q)\ \forall q' \in X\}$ for $q \in \operatorname{dom}\Psi$. For a convex function $\gamma$ on $[0,\infty)$, $\gamma'_\pm$ denote its one-sided derivatives. Weak convergence is denoted by $\rightharpoonup$.

## 2. The method at the continuous level

### 2.1. Setting.

**Definition 2.1** (Young integrand)**.** A function $G\colon \Omega \times [0,\infty) \to [0,\infty)$ is a *Young integrand* if $G(\cdot,t)$ is measurable for every $t \geq 0$ and, for a.e. $x \in \Omega$, $G(x,\cdot)$ is finite, convex and nondecreasing with $G(x,0) = 0$.

A finite convex nondecreasing function on $[0,\infty)$ vanishing at 0 is continuous (at 0: $0 \leq G(x,t) \leq (t/s)G(x,s)$ for $0 < t < s$), so $G$ is a Carathéodory function and $x \mapsto G(x,|q(x)|)$ is measurable for every measurable $q$. We write

$$\Phi(x,\xi) = G(x,|\xi|), \qquad g(x,t) = \partial_t^+ G(x,t) \quad (t \geq 0),$$

so that $\Phi(x,\cdot)$ is convex on $\mathbb{R}^d$ (composition of a convex nondecreasing function with a norm) and $g(x,\cdot)$ is nondecreasing, finite and nonnegative; $g(x,0) = \inf_{s>0} G(x,s)/s$. No growth condition ($\Delta_2$, $\nabla_2$, superlinearity) is assumed in this section.

Let $V = H_0^1(\Omega)$ with the norm $\|v\|_V = \|\nabla v\|$, $H = L^2(\Omega;\mathbb{R}^d)$, $f \in L^2(\Omega)$, and

$$F(q) = \int_\Omega \Phi(x, q(x))\,dx \in [0,+\infty], \qquad q \in H; \qquad J(v) = F(\nabla v) - (f,v), \quad v \in V.$$

**Lemma 2.2.** *$F$ is convex, $F \geq 0 = F(0)$, and $F$ is lower semicontinuous on $H$; hence also weakly lower semicontinuous.*

*Proof.* Convexity and $F \geq 0 = F(0)$ are clear. If $q_k \to q$ in $H$, choose a subsequence with $F(q_{k_j}) \to \liminf_k F(q_k)$ and $q_{k_j} \to q$ a.e.; by continuity of $\Phi(x,\cdot)$ and Fatou's lemma $F(q) \leq \liminf_j F(q_{k_j})$. The sublevel sets of $F$ are convex and closed, hence weakly closed (Mazur), which is weak lower semicontinuity. □

Abstract structure. We write $B = \nabla\colon V \to H$ and $\ell(v) = (f, v)$. Sections 2.2–2.4 use only:

(A1) $V$ and $H$ are real Hilbert spaces and $B\colon V \to H$ is linear with $\|Bv\|_H = \|v\|_V$;
(A2) $F\colon H \to [0, +\infty]$ is convex, lower semicontinuous and proper ($\operatorname{dom} F \neq \emptyset$);
(A3) $\ell \in V^*$.

In the concrete case we also use the compactness of $H_0^1(\Omega) \hookrightarrow L^2(\Omega)$ (Rellich's theorem, valid for every bounded open set) and the pointwise description of $\partial F$ (Lemma 2.4). Section 4 uses the same structure for the finite element problem.

2.2. **Augmented Lagrangian and saddle points.** For $r > 0$ define on $V \times H \times H$

$$\mathcal{L}_r(v, q, \mu) = F(q) - \ell(v) + (\mu, Bv - q) + \frac{r}{2}\|Bv - q\|^2.$$

A triple $(u^*, \nu^*, \eta^*)$ is a saddle point if $\mathcal{L}_r(u^*, \nu^*, \mu) \leq \mathcal{L}_r(u^*, \nu^*, \eta^*) \leq \mathcal{L}_r(v, q, \eta^*)$ for all $(v, q, \mu)$, with the middle term finite.

**Proposition 2.3** (Saddle points)**.** *Assume (A1)–(A3). $(u^*, \nu^*, \eta^*) \in V \times H \times H$ is a saddle point of $\mathcal{L}_r$ if and only if*

$$\nu^* = Bu^*, \qquad \eta^* \in \partial F(\nu^*), \qquad (\eta^*, Bw) = \ell(w) \quad \forall w \in V. \tag{2.1}$$

*In that case $F(\nu^*) < \infty$ and $u^*$ minimises $J(v) = F(Bv) - \ell(v)$ over $V$. In particular the set of saddle points does not depend on $r$.*

*Proof.* ($\Leftarrow$) Since $Bu^* = \nu^*$, $\mathcal{L}_r(u^*, \nu^*, \mu) = F(\nu^*) - \ell(u^*)$ does not depend on $\mu$, and it is finite because $\eta^* \in \partial F(\nu^*)$ requires $\nu^* \in \operatorname{dom} F$. For $(v, q) \in V \times H$, the subgradient inequality $F(q) \geq F(\nu^*) + (\eta^*, q - \nu^*)$ and (2.1) with $w = v - u^*$ give

$$\begin{aligned}\mathcal{L}_r(v, q, \eta^*) - \mathcal{L}_r(u^*, \nu^*, \eta^*) &\geq (\eta^*, q - \nu^*) - \ell(v - u^*) + (\eta^*, Bv - q) + \frac{r}{2}\|Bv - q\|^2 \\ &= (\eta^*, B(v - u^*)) - \ell(v - u^*) + \frac{r}{2}\|Bv - q\|^2 \geq 0,\end{aligned}$$

where we used $\nu^* = Bu^*$.

($\Rightarrow$) The middle value is finite, so $F(\nu^*) < \infty$. The left inequality, $(\mu, Bu^* - \nu^*) \leq (\eta^*, Bu^* - \nu^*)$ for all $\mu \in H$, forces $Bu^* = \nu^*$. The right inequality with $(v, q) = (u^* + sw, \nu^*)$, $w \in V$, $s \in \mathbb{R}$, reads $s\big[(\eta^*, Bw) - \ell(w)\big] + \frac{rs^2}{2}\|Bw\|^2 \geq 0$; dividing by $s > 0$ and by $s < 0$ and letting $s \to 0$ gives $(\eta^*, Bw) = \ell(w)$. The right inequality with $v = u^*$ and $q = \nu^* + s(q' - \nu^*)$, $q' \in \operatorname{dom} F$, $s \in (0, 1)$, together with convexity, gives

$$F(\nu^*) \leq F(\nu^*) + s\big(F(q') - F(\nu^*)\big) - s(\eta^*, q' - \nu^*) + \frac{rs^2}{2}\|q' - \nu^*\|^2;$$

dividing by $s$ and letting $s \downarrow 0$ yields $F(q') \geq F(\nu^*) + (\eta^*, q' - \nu^*)$, trivially true also if $q' \notin \operatorname{dom} F$. Hence $\eta^* \in \partial F(\nu^*)$.

Finally, if (2.1) holds, then for $v \in V$: $J(v) - J(u^*) \geq (\eta^*, B(v - u^*)) - \ell(v - u^*) = 0$. Condition (2.1) does not involve $r$. □

**Assumption S.** The augmented Lagrangian $\mathcal{L}_r$ has a saddle point in $V \times H \times H$; equivalently, (2.1) has a solution.

In the concrete case, (S) says that $J$ has a minimiser $u \in H^1_0(\Omega)$ whose *flux* $\eta \in \partial F(\nabla u)$ (Lemma 2.4) belongs to $L^2(\Omega;\mathbb{R}^d)$ and satisfies $(\eta, \nabla w) = (f, w)$ for all $w \in H^1_0$. It is a property of the solution, not only of $G$ (Section 3).

**Lemma 2.4** (Pointwise description of $\partial F$)**.** *Let $q \in \operatorname{dom} F$ and $\eta \in H$. Then $\eta \in \partial F(q)$ if and only if $\eta(x) \in \partial\Phi(x,\cdot)(q(x))$ for a.e. $x \in \Omega$.*

*Proof.* ($\Leftarrow$) If $q' \in \operatorname{dom} F$, the pointwise inequality $\Phi(x,q'(x)) \geq \Phi(x,q(x)) + \eta(x)\cdot(q'(x)-q(x))$ involves integrable functions and can be integrated; if $q' \notin \operatorname{dom} F$ there is nothing to prove.

($\Rightarrow$) Fix $z \in \mathbb{Q}^d$ and $M \in \mathbb{N}$, let $\Omega^z_M = \{x : \Phi(x,z) \leq M\}$ and let $A \subset \Omega^z_M$ be measurable. The field $q' = q\,\mathbf{1}_{\Omega\setminus A} + z\,\mathbf{1}_A$ belongs to $H \cap \operatorname{dom} F$, and $F(q') \geq F(q) + (\eta, q'-q)$ reads

$$\int_A \big[\Phi(x,z) - \Phi(x,q(x)) - \eta(x)\cdot(z-q(x))\big]\,dx \geq 0.$$

The integrand is integrable on $\Omega^z_M$ and $A$ is arbitrary, so it is $\geq 0$ a.e. on $\Omega^z_M$. Since $\Phi(x,z) < \infty$, $\bigcup_M \Omega^z_M = \Omega$ up to a null set. Taking the union of the exceptional null sets over $z \in \mathbb{Q}^d$ we find a null set $N$ such that $\Phi(x,z) \geq \Phi(x,q(x)) + \eta(x)\cdot(z-q(x))$ for $x \notin N$ and all $z \in \mathbb{Q}^d$, hence, by continuity of $\Phi(x,\cdot)$, for all $z \in \mathbb{R}^d$. □

**Lemma 2.5** (Subdifferential of $\Phi$)**.** *Let $\gamma\colon [0,\infty) \to [0,\infty)$ be finite, convex, nondecreasing, $\gamma(0) = 0$, and $\phi(\xi) = \gamma(|\xi|)$ on $\mathbb{R}^d$.*

(i) *$\partial\phi(0) = \{\zeta : |\zeta| \leq \gamma'_+(0)\}$.*
(ii) *If $\xi \neq 0$ and $\gamma$ is differentiable at $|\xi|$, then $\partial\phi(\xi) = \{\gamma'(|\xi|)\xi/|\xi|\}$.*
(iii) *If $\gamma \in C^1([0,\infty))$ with $\gamma'(0) = 0$, then $\phi \in C^1(\mathbb{R}^d)$, $\nabla\phi(\xi) = T(\xi) := \gamma'(|\xi|)\xi/|\xi|$ ($T(0) = 0$), and $\partial\phi(\xi) = \{T(\xi)\}$ for every $\xi$.*

*Proof.* (i) $\zeta \in \partial\phi(0)$ iff $\gamma(|z|) \geq \zeta\cdot z$ for all $z$, iff $\gamma(t) \geq |\zeta| t$ for all $t \geq 0$, iff $|\zeta| \leq \inf_{t>0}\gamma(t)/t = \gamma'_+(0)$ (the difference quotient of a convex function is nondecreasing). (ii) and (iii): a convex function differentiable at $\xi$ has $\partial\phi(\xi) = \{\nabla\phi(\xi)\}$: the gradient is a subgradient by convexity, and if $\zeta \in \partial\phi(\xi)$ then $(\phi(\xi+sz)-\phi(\xi))/s \geq \zeta\cdot z$ for $s>0$, so $\nabla\phi(\xi)\cdot z \geq \zeta\cdot z$ for all $z$, i.e. $\zeta = \nabla\phi(\xi)$. At $\xi \neq 0$ the chain rule gives $\nabla\phi(\xi) = \gamma'(|\xi|)\xi/|\xi|$. In (iii), $\phi(\xi) = \gamma(|\xi|) = o(|\xi|)$ at 0 because $\gamma(0) = \gamma'(0) = 0$, so $\nabla\phi(0) = 0$, and $T$ is continuous at 0 because $\gamma'(t) \to 0$. □

In particular, when $G(x,\cdot) \in C^1([0,\infty))$ with $g(x,0) = 0$, Lemmas 2.4 and 2.5 give, for $q \in \operatorname{dom} F$,

$$\partial F(q) = \{T(\cdot,q)\} \ \text{ if } T(\cdot,q) \in H, \qquad \partial F(q) = \emptyset \ \text{ otherwise;} \qquad T(x,\xi) = g(x,|\xi|)\frac{\xi}{|\xi|}. \tag{2.2}$$

For $G(x,t) = e^t - 1$, instead, $g(x,0) = 1$ and $\partial\Phi(x,0)$ is the closed unit ball: $\partial F$ is multivalued at fields that vanish on a set of positive measure.

2.3. **The algorithm.** Fix $r > 0$, $\nu_0 \in H$ and $\eta_1 \in \partial F(\nu_0)$. If $F(0) = 0$, as in Section 2.1, then $0 \in \partial F(0)$ because $F \geq 0$, so $\nu_0 = \eta_1 = 0$ (the initialisation used in the code) is admissible; by Remark 2.9 the condition $\eta_1 \in \partial F(\nu_0)$ is in fact not needed. For $n \geq 1$:

$$u_n \in V: \quad (\eta_n + r(Bu_n - \nu_{n-1}), Bw) = \ell(w) \quad \forall w \in V, \tag{2.3}$$

$$\nu_n = \operatorname*{argmin}_{q\in H}\Big\{F(q) - (\eta_n, q) + \frac{r}{2}\|Bu_n - q\|^2\Big\}, \tag{2.4}$$

$$\eta_{n+1} = \eta_n + r(Bu_n - \nu_n). \tag{2.5}$$

Step (2.3) minimises $\mathcal{L}_r(\cdot,\nu_{n-1},\eta_n)$, step (2.4) minimises $\mathcal{L}_r(u_n,\cdot,\eta_n)$, and (2.5) is a multiplier step of length $\rho = r$. This is ALG2 of [36], [39, Ch. VI, (3.7)–(3.10)] with $\rho = r$ (same order of the steps;

in [41] the $\nu$-step comes first), as used in [4]. In the concrete case (2.3) is the Poisson problem $-r\Delta u_n = f + \operatorname{div}(\eta_n - r\nu_{n-1})$, $u_n \in H^1_0$, with the same operator at every iteration. We write

$$s_n = Bu_n - \nu_n \qquad \text{(primal residual)}, \qquad \text{so that} \quad \eta_{n+1} - \eta_n = r s_n.$$

**Lemma 2.6** (Well-posedness)**.** *Assume (A1)–(A3).*

(i) (2.3) *has a unique solution* $u_n$.

(ii) (2.4) *has a unique solution, and* $\nu \in H$ *is that solution if and only if* $\eta_n + r(Bu_n - \nu) \in \partial F(\nu)$. *Consequently*

$$\eta_{n+1} \in \partial F(\nu_n) \qquad (n \geq 0). \tag{2.6}$$

(iii) *For* $n \geq 1$,

$$(\eta_{n+1} + r(\nu_n - \nu_{n-1}), Bw) = \ell(w) \qquad \forall w \in V. \tag{2.7}$$

*Proof.* (i) By (A1), $(v, w) \mapsto r(Bv, Bw)$ is $r$ times the inner product of $V$, and $w \mapsto \ell(w) - (\eta_n - r\nu_{n-1}, Bw)$ is in $V^*$; apply the Riesz theorem.

(ii) Let $\Psi_n(q) = F(q) - (\eta_n, q) + \frac{r}{2}\|Bu_n - q\|^2$. It is proper (since $\operatorname{dom} F \neq \emptyset$), convex, lower semicontinuous, hence weakly lower semicontinuous, and coercive: $\Psi_n(q) \geq -\|\eta_n\|\|q\| + \frac{r}{2}(\|q\| - \|Bu_n\|)^2$ because $F \geq 0$. Bounded sets are weakly sequentially compact, so a minimiser exists; it is unique because $\Psi_n$ is strictly convex. If $\nu$ minimises $\Psi_n$, then for $q \in \operatorname{dom} F$ and $s \in (0,1)$, with $\nu_s = \nu + s(q - \nu)$,

$$0 \leq \Psi_n(\nu_s) - \Psi_n(\nu) \leq s\big(F(q) - F(\nu)\big) - s(\eta_n, q - \nu) - rs(Bu_n - \nu, q - \nu) + \frac{rs^2}{2}\|q - \nu\|^2;$$

dividing by $s$ and letting $s \downarrow 0$ gives $F(q) \geq F(\nu) + (\eta_n + r(Bu_n - \nu), q - \nu)$, i.e. $\eta_n + r(Bu_n - \nu) \in \partial F(\nu)$. Conversely, if this inclusion holds, adding the subgradient inequality to the identity $\frac{r}{2}\|Bu_n - q\|^2 - \frac{r}{2}\|Bu_n - \nu\|^2 = -r(Bu_n - \nu, q - \nu) + \frac{r}{2}\|q - \nu\|^2$ shows $\Psi_n(q) \geq \Psi_n(\nu)$. With $\nu = \nu_n$, (2.5) gives (2.6) for $n \geq 1$; for $n = 0$ it is the choice of $\eta_1$.

(iii) By (2.5), $\eta_n + r(Bu_n - \nu_{n-1}) = \eta_{n+1} + r(\nu_n - \nu_{n-1})$; insert in (2.3). □

The second step decouples into independent pointwise problems, described by the following lemma; the code solves them by bisection for a scalar monotone equation.

**Lemma 2.7** (The pointwise problem)**.** *Let* $\gamma$ *be as in Lemma 2.5,* $\phi(\xi) = \gamma(|\xi|)$, $r > 0$ *and* $w \in \mathbb{R}^d$.

(i) $\xi \mapsto \phi(\xi) + \frac{r}{2}|\xi - w|^2$ *has a unique minimiser* $\hat\xi = \hat t\, w/|w|$ ($\hat\xi = 0$ *if* $w = 0$), *where* $\hat t \in [0, |w|]$ *is the unique minimiser of* $\chi(t) = \gamma(t) + \frac{r}{2}(t - |w|)^2$ *on* $[0, \infty)$.

(ii) $\hat t = 0$ *if and only if* $r|w| \leq \gamma'_+(0)$; *otherwise* $\hat t > 0$ *is characterised by* $\gamma'_-(\hat t) \leq r(|w| - \hat t) \leq \gamma'_+(\hat t)$.

(iii) *If* $\gamma \in C^1$ *and* $\gamma'(0) < r|w|$, *then* $\hat t$ *is the unique root in* $(0, |w|]$ *of the strictly increasing function* $t \mapsto \gamma'(t) + rt - r|w|$.

(iv) $\hat\eta := r(w - \hat\xi) \in \partial\phi(\hat\xi)$, *and* $|\hat\eta| = r(|w| - \hat t)$.

*Proof.* (i) The function $\chi$ is continuous, strictly convex and coercive on $[0, \infty)$, so it has a unique minimiser $\hat t$; for $t > |w|$, $\chi(t) > \chi(|w|)$ because $\gamma$ is nondecreasing, so $\hat t \leq |w|$. For every $\xi$, $|\xi - w| \geq \big||\xi| - |w|\big|$, hence $\phi(\xi) + \frac{r}{2}|\xi - w|^2 \geq \chi(|\xi|) \geq \chi(\hat t)$, and the value $\chi(\hat t)$ is attained at $\hat\xi$. Uniqueness follows from strict convexity. (ii) For a convex function on $[0, \infty)$, $\hat t$ is a minimiser iff $\chi'_+(\hat t) \geq 0$ and, if $\hat t > 0$, $\chi'_-(\hat t) \leq 0$; and $\chi'_\pm(t) = \gamma'_\pm(t) + r(t - |w|)$. (iii) follows from (ii) and the strict monotonicity of $\gamma'(t) + rt$. (iv) If $w = 0$, $\hat\eta = 0 \in \partial\phi(0)$. If $\hat t = 0 < |w|$, $|\hat\eta| = r|w| \leq \gamma'_+(0)$ and Lemma 2.5(i) applies. If $\hat t > 0$, then for every $z$, using $|z| \geq \hat w \cdot z$ with $\hat w = w/|w|$, the monotonicity of $\gamma$ and the convexity inequality $\gamma(s) \geq \gamma(\hat t) + \lambda(s - \hat t)$ with $\lambda = r(|w| - \hat t) \in [\gamma'_-(\hat t), \gamma'_+(\hat t)]$ (valid for $s \geq 0$),

$$\phi(z) \geq \gamma(\max\{\hat w \cdot z, 0\}) \geq \gamma(\hat t) + \lambda\big(\hat w \cdot z - \hat t\big) = \phi(\hat\xi) + \hat\eta \cdot (z - \hat\xi),$$

where in the case $\hat{w}\cdot z<0$ we used $\gamma(0)=0\geq\gamma(\hat{t})+\lambda(\hat{w}\cdot z-\hat{t})$, which follows from the convexity inequality at $s=0$ and $\lambda\geq 0$. □

**Corollary 2.8** (Pointwise form of (2.4))**.** *In the concrete case, let $w_n=\eta_n/r+\nabla u_n$. For a.e. $x$, $\nu_n(x)$ is the minimiser of Lemma 2.7 with $\gamma=G(x,\cdot)$ and $w=w_n(x)$: $\nu_n(x)=t(x)\,w_n(x)/|w_n(x)|$, where $t(x)=0$ if $r|w_n(x)|\leq g(x,0)$ and otherwise, when $G(x,\cdot)\in C^1$, $t(x)\in(0,|w_n(x)|]$ solves*

$$g(x,t)+r\,t=r|w_n(x)|. \tag{2.8}$$

*Moreover $\eta_{n+1}(x)=r(w_n(x)-\nu_n(x))\in\partial\Phi(x,\cdot)(\nu_n(x))$; when $\nu_n(x)\neq 0$ and $G(x,\cdot)$ is differentiable at $|\nu_n(x)|$, $\eta_{n+1}(x)=g(x,|\nu_n(x)|)\nu_n(x)/|\nu_n(x)|$: the multiplier is the flux.*

*Proof.* By (2.6) and Lemma 2.4, $\zeta(x):=\eta_{n+1}(x)=r(w_n(x)-\nu_n(x))\in\partial\Phi(x,\cdot)(\nu_n(x))$ a.e. Adding the subgradient inequality to $\frac{r}{2}|z-w|^2-\frac{r}{2}|\xi-w|^2\geq r(\xi-w)\cdot(z-\xi)$ shows that $\xi=\nu_n(x)$ minimises $\Phi(x,\cdot)+\frac{r}{2}|\cdot-w_n(x)|^2$; by uniqueness it is the minimiser of Lemma 2.7. The rest is Lemma 2.7 and Lemma 2.5(ii). □

### 2.4. **Theorem A: weak convergence under (S).**

**Theorem A.** *Assume (A1)–(A3) and Assumption S. Let $(u_n,\nu_n,\eta_n)$ be generated by (2.3)–(2.5), let $(u^*,\nu^*,\eta^*)$ be any saddle point, and set*

$$E_n=\frac{1}{2r}\|\eta_{n+1}-\eta^*\|^2+\frac{r}{2}\|\nu_n-\nu^*\|^2\qquad(n\geq 0).$$

*Then for every $n\geq 1$*

$$E_n+\frac{r}{2}\|s_n\|^2+\frac{r}{2}\|\nu_n-\nu_{n-1}\|^2+(\eta_{n+1}-\eta^*,\nu_n-\nu^*)+r(s_n,\nu_n-\nu_{n-1})=E_{n-1}, \tag{2.9}$$

*and the last two terms on the left are nonnegative. Consequently $(E_n)$ is nonincreasing; $(\eta_n)$, $(\nu_n)$ are bounded in $H$ and $(u_n)$ in $V$;*

$$\sum_{n\geq 1}\|s_n\|^2<\infty,\qquad\sum_{n\geq 1}\|\nu_n-\nu_{n-1}\|^2<\infty,\qquad\sum_{n\geq 1}m_n<\infty,\quad m_n:=(\eta_{n+1}-\eta^*,\nu_n-\nu^*)\geq 0;$$

*in particular $Bu_n-\nu_n\to 0$, $\nu_n-\nu_{n-1}\to 0$ and $\eta_{n+1}-\eta_n\to 0$ strongly in $H$. Moreover there is a saddle point $(\bar{u},\bar{\nu},\bar{\eta})$ such that*

$$u_n\rightharpoonup\bar{u}\ \text{ in } V,\qquad\nu_n\rightharpoonup\bar{\nu},\quad\eta_n\rightharpoonup\bar{\eta}\ \text{ in } H.$$

*In the concrete case $V=H_0^1(\Omega)$: $u_n\rightharpoonup\bar{u}$ weakly in $H_0^1(\Omega)$ and strongly in $L^2(\Omega)$, $\nabla u_n-\nu_n\to 0$ in $L^2(\Omega;\mathbb{R}^d)$, and $\bar{u}$ minimises $J$ over $H_0^1(\Omega)$; if this minimiser is unique (for instance if $G(x,\cdot)$ is strictly convex for a.e. $x$), $\bar{u}$ is the unique minimiser.*

No growth condition and no differentiability of $G$ is used: whenever (S) holds, Theorem A applies to $t^p/p$ for every $p>1$, to $p(x)$, $p$-$q$ and two-phase energies, to $t^p\log^a(1+t)$, to $e^t-1-t$ and to $e^t-1$ (where $\partial F$ is multivalued); for the uniformly convex ones among them Glowinski's theorem already gives strong convergence (Section 2.6).

*Proof. Step 1: the identity (2.9).* Fix $n\geq 1$. Subtract $(\eta^*,Bw)=\ell(w)$ from (2.7) and take $w=u_n-u^*$; since $\nu^*=Bu^*$, $Bw=s_n+(\nu_n-\nu^*)$, and we obtain

$$(\eta_{n+1}-\eta^*,s_n)+(\eta_{n+1}-\eta^*,\nu_n-\nu^*)+r(\nu_n-\nu_{n-1},s_n)+r(\nu_n-\nu_{n-1},\nu_n-\nu^*)=0.$$

By the identity $2(a-c,a-b)=\|a-c\|^2+\|a-b\|^2-\|b-c\|^2$ and $rs_n=\eta_{n+1}-\eta_n$,

$$(\eta_{n+1}-\eta^*,s_n)=\frac{1}{r}(\eta_{n+1}-\eta^*,\eta_{n+1}-\eta_n)=\frac{1}{2r}\big(\|\eta_{n+1}-\eta^*\|^2-\|\eta_n-\eta^*\|^2\big)+\frac{r}{2}\|s_n\|^2,$$
$$r(\nu_n-\nu_{n-1},\nu_n-\nu^*)=\frac{r}{2}\big(\|\nu_n-\nu^*\|^2-\|\nu_{n-1}-\nu^*\|^2\big)+\frac{r}{2}\|\nu_n-\nu_{n-1}\|^2.$$

Inserting these two identities gives (2.9). Note that Step 1 uses only (2.7), $\nu^* = Bu^*$ and $(\eta^*, Bw) = \ell(w)$.

*Step 2: signs and consequences.* By (2.6), (2.1) and the monotonicity of $\partial F$ (add the two subgradient inequalities), $m_n = (\eta_{n+1} - \eta^*, \nu_n - \nu^*) \geq 0$. Also $\eta_n \in \partial F(\nu_{n-1})$ for $n \geq 1$, so $r(s_n, \nu_n - \nu_{n-1}) = (\eta_{n+1} - \eta_n, \nu_n - \nu_{n-1}) \geq 0$. Thus $E_n \leq E_{n-1}$, and summing (2.9) from 1 to $N$ gives $E_N + \sum_{n=1}^{N} \left(\frac{r}{2}\|s_n\|^2 + \frac{r}{2}\|\nu_n - \nu_{n-1}\|^2 + m_n\right) \leq E_0$. Hence $(\eta_{n+1})$ and $(\nu_n)$ are bounded, the three series converge, and $Bu_n = s_n + \nu_n$ is bounded, i.e. $(u_n)$ is bounded in $V$ by (A1).

*Step 3: weak cluster points are saddle points.* Let $x_n = (\nu_n, \eta_{n+1}) \in H \times H$ and let $(\bar\nu, \bar\eta)$ be a weak cluster point: $\nu_{n_k} \rightharpoonup \bar\nu$, $\eta_{n_k+1} \rightharpoonup \bar\eta$; passing to a further subsequence, $u_{n_k} \rightharpoonup \bar u$ in $V$. Since $B$ is linear and bounded, $Bu_{n_k} \rightharpoonup B\bar u$; since $Bu_n - \nu_n \to 0$, $B\bar u = \bar\nu$. In (2.7) with $n = n_k$ the term $r(\nu_n - \nu_{n-1}, Bw)$ tends to 0, so $(\bar\eta, Bw) = \ell(w)$ for all $w \in V$. Taking $w = u_n$ in (2.7) and using $Bu_n = s_n + \nu_n$,

$$(\eta_{n+1}, \nu_n) = \ell(u_n) - (\eta_{n+1}, s_n) - r(\nu_n - \nu_{n-1}, Bu_n) \xrightarrow[n=n_k\to\infty]{} \ell(\bar u) = (\bar\eta, B\bar u) = (\bar\eta, \bar\nu),$$

because the last two terms are products of a bounded and a strongly null sequence and $\ell(u_{n_k}) \to \ell(\bar u)$. Now, by (2.6), for every $q \in H$,

$$F(q) \geq F(\nu_{n_k}) + (\eta_{n_k+1}, q) - (\eta_{n_k+1}, \nu_{n_k}).$$

Letting $k \to \infty$ and using the weak lower semicontinuity of $F$ (Lemma 2.2 in the concrete case; in general (A2) and Mazur's lemma) we get $F(q) \geq F(\bar\nu) + (\bar\eta, q - \bar\nu)$. Taking $q = \nu^*$ shows $F(\bar\nu) < \infty$, so $\bar\eta \in \partial F(\bar\nu)$. By Proposition 2.3, $(\bar u, \bar\nu, \bar\eta)$ is a saddle point.

*Step 4: convergence of the whole sequence.* Let $\mathcal{S}$ be the set of pairs $(\nu^*, \eta^*)$ such that $(u^*, \nu^*, \eta^*)$ is a saddle point for some $u^* \in V$, and equip $H \times H$ with the inner product $\langle(\nu, \eta), (\nu', \eta')\rangle = r(\nu, \nu') + \frac{1}{r}(\eta, \eta')$ and norm $|\cdot|$. Then $|x_n - z|^2 = 2E_n(z)$ for $z \in \mathcal{S}$, which converges by Step 2, and every weak cluster point of the bounded sequence $(x_n)$ lies in $\mathcal{S}$ by Step 3. Let $z_1, z_2$ be two weak cluster points. Since

$$|x_n - z_1|^2 - |x_n - z_2|^2 = -2\langle x_n, z_1 - z_2\rangle + |z_1|^2 - |z_2|^2$$

converges, $\langle x_n, z_1 - z_2\rangle$ converges; evaluating the limit along the two subsequences gives $\langle z_1, z_1 - z_2\rangle = \langle z_2, z_1 - z_2\rangle$, i.e. $|z_1 - z_2|^2 = 0$. Hence $x_n \rightharpoonup \bar z = (\bar\nu, \bar\eta) \in \mathcal{S}$, and $\eta_n \rightharpoonup \bar\eta$ since $\eta_{n+1} - \eta_n \to 0$. Finally $Bu_n = s_n + \nu_n \rightharpoonup \bar\nu$; every weak cluster point $\bar u$ of the bounded sequence $(u_n)$ satisfies $B\bar u = \bar\nu$, and $B$ is injective, so $u_n \rightharpoonup \bar u$ in $V$, and $(\bar u, \bar\nu, \bar\eta)$ is a saddle point by Step 3.

In the concrete case, Rellich's theorem upgrades the weak convergence in $H_0^1(\Omega)$ to strong convergence in $L^2(\Omega)$, and Proposition 2.3 shows that $\bar u$ minimises $J$. □

*Remark* 2.9 (On the initialisation). Since $s_n + (\nu_n - \nu_{n-1}) = Bu_n - \nu_{n-1}$, (2.9) reads $E_n + m_n + \frac{r}{2}\|Bu_n - \nu_{n-1}\|^2 = E_{n-1}$ for every $n \geq 1$, whatever $\eta_1$ is. Hence $E_n$ is nonincreasing without the condition $\eta_1 \in \partial F(\nu_0)$, which is used only to split the summability of $\|Bu_n - \nu_{n-1}\|^2$ into that of $\|s_n\|^2$ and $\|\nu_n - \nu_{n-1}\|^2$ at $n = 1$; all conclusions of Theorem A remain true.

*Remark* 2.10 (Related results; weak convergence only). Identity (2.9) is the equality version, for $\rho = r$, of Glowinski's inequality [39, Ch. VI, (5.25)]: dividing (5.25) by $2r$ gives $E_{n-1} - E_n \geq m_n + \frac{r}{2}\|s_n\|^2 + \frac{r}{2}\|\nu_n - \nu_{n-1}\|^2$ (there with $F_0'$ in $m_n$), and the weak convergence of the whole multiplier sequence via Opial's lemma is stated in [39, Ch. VI, §7]. Theorem A shows that (iii)–(iv) of Section 2.6 are not needed for weak convergence. The same functional appears in [11, Thm. 3.7] and [14, App. A]. Weak convergence also follows from the equivalence of ADMM with Douglas–Rachford splitting for the dual [37], [31, §5] and the weak convergence of the latter [46, 51, 12]; see [49]. The example of Bùi and Combettes [16] ($\mathcal{A} = N_{\{e_0\}^\perp}$, $\mathcal{B} = \partial\psi$, $\psi$ convex), in which the Douglas–Rachford shadow sequence converges weakly but not strongly, is, through the correspondence of Proposition 2.11, an instance of (A1)–(A3) ($V = \mathbb{R}$, $Bt = te_0$, $\ell = 0$, $F = \psi^* - \psi^*(0)$, $r = 1$) in which the shadow sequence is the multiplier $\eta_n$, while $\nu_n \to 0$ and $u_n$ converge strongly. Thus

*the weak convergence of $\eta_n$ in Theorem A cannot be improved under (A1)–(A3)*; the example says nothing about $\nu_n$ or $u_n$, and we do not know whether strong convergence of $u_n$ can fail for some instance of problem (1.1).

**Proposition 2.11** ((S) is necessary for boundedness)**.** *Assume (A1)–(A3) and let $(u_n, \nu_n, \eta_n)$ be generated by (2.3)–(2.5) from any $\nu_0, \eta_1 \in H$. Then* (S) *holds if and only if $(\eta_n + r\nu_{n-1})_n$ is bounded in $H$. In particular, if* (S) *fails, $(\eta_n)$ or $(\nu_n)$ is unbounded.*

*Proof.* If (S) holds, $(\eta_n)$ and $(\nu_n)$ are bounded by Theorem A (with Remark 2.9 if $\eta_1 \notin \partial F(\nu_0)$). Conversely, (2.3)–(2.5) is the alternating direction method of multipliers of [31, Thm. 8] for $\min f(v) + F(Bv)$, $f = -\ell$, with penalty $r$, relaxation 1 and exact steps. The proof there (relations (Y1)–(Y3)) is algebraic and holds verbatim in Hilbert space: $z_n := \eta_n + r\nu_{n-1}$ is the Douglas–Rachford sequence for the maximal monotone operators $\mathcal{A} = \partial(\iota_{\{B^*\eta = \ell\}})$ and $\mathcal{B} = \partial F^*$, with $\eta_n = J_{r\mathcal{B}}(z_n)$ for $n \geq 2$ (an initial error at $n = 1$ is allowed there). $\{B^*\eta = \ell\}$ is a nonempty closed affine subspace, since $B^*$ is onto by (A1), and its normal cone is $R(B)$. If $0 \in \mathcal{A}\eta + \mathcal{B}\eta$, there is $a \in \partial F^*(\eta)$ with $-a \in R(B)$, i.e. $a = Bu$, $\eta \in \partial F(Bu)$ and $B^*\eta = \ell$: this is (2.1). Hence if (S) fails, $\operatorname{zer}(\mathcal{A} + \mathcal{B}) = \emptyset$ and $(z_n)$ is unbounded by [31, Thm. 7], which is stated in a general Hilbert space. □

2.5. **Theorem B: strong convergence.** Assume now that $G(x, \cdot) \in C^1([0, \infty))$ with $g(x, 0) = 0$ for a.e. $x$, and let $T(x, \xi) = g(x, |\xi|)\xi/|\xi|$ and

$$k(x,t) = \frac{g(x,t)}{t} \ (t > 0), \qquad \psi(x; a, b) = (T(x,a) - T(x,b)) \cdot (a - b) \geq 0.$$

By (2.2) and (2.6), $\eta_{n+1} = T(\cdot, \nu_n)$ and $\eta^* = T(\cdot, \nu^*)$ a.e. for every saddle point, hence

$$m_n = \int_\Omega \psi(x; \nu_n(x), \nu^*(x))\, dx. \tag{2.10}$$

**Lemma 2.12.** (i) *If $t \mapsto k(x,t)$ is nondecreasing on $(0, \infty)$ and we set $k(x, 0) = \lim_{t \downarrow 0} k(x,t)$, then for all $a, b \in \mathbb{R}^d$*

$$\psi(x; a, b) = \tfrac{1}{2}\big(k(x,|a|) + k(x,|b|)\big)|a-b|^2 + \tfrac{1}{2}\big(k(x,|a|) - k(x,|b|)\big)\big(|a|^2 - |b|^2\big) \ \geq \ \tfrac{1}{2}\, k\big(x, \tfrac{|a-b|}{2}\big)|a-b|^2.$$

(ii) *Let $T(\xi) = |\xi|^{p-2}\xi$ with $1 < p \leq 2$. Then for all $a, b \in \mathbb{R}^d$*

$$(T(a) - T(b)) \cdot (a - b) \geq (p-1)(|a| + |b|)^{p-2}|a-b|^2 \quad (a, b \text{ not both } 0), \qquad |T(a) - T(b)| \leq 2^p |a-b|^{p-1}.$$

*Proof.* (i) Write $k_a = k(x, |a|)$, $k_b = k(x, |b|)$, so $T(x, a) = k_a a$ (also for $a = 0$). Expanding, both $\psi$ and the right-hand side of the identity equal $k_a|a|^2 + k_b|b|^2 - (k_a + k_b)\, a \cdot b$. The second term is $\geq 0$ because $k(x, \cdot)$ is nondecreasing on $[0, \infty)$. Hence $\psi \geq \frac{1}{2}(k_a + k_b)|a-b|^2 \geq \frac{1}{2} k(x, \max\{|a|, |b|\})|a-b|^2$, and $\max\{|a|, |b|\} \geq |a - b|/2$.

(ii) $T$ is $C^1$ on $\mathbb{R}^d \setminus \{0\}$ with $DT(\xi) = |\xi|^{p-2}\big(I + (p-2)\hat{\xi} \otimes \hat{\xi}\big)$, $\hat{\xi} = \xi/|\xi|$, a symmetric matrix with eigenvalues $|\xi|^{p-2}$ and $(p-1)|\xi|^{p-2}$. Let $c_s = b + s(a - b)$, $s \in [0, 1]$. The segment meets 0 at most at one $s_0$ (unless $a = b$, a trivial case), $s \mapsto T(c_s)$ is continuous on $[0, 1]$, $C^1$ off $s_0$, and $|\frac{d}{ds} T(c_s)| \leq |c_s|^{p-2}|a - b|$, which is integrable because $|c_s| \geq |a - b||s - s_0|$ if the segment meets 0 and $p - 2 > -1$. Hence $s \mapsto T(c_s)$ is absolutely continuous and

$$\begin{aligned}(T(a) - T(b)) \cdot (a - b) &= \int_0^1 (a-b)^\top DT(c_s)(a-b)\, ds \\ &\geq (p-1)|a-b|^2 \int_0^1 |c_s|^{p-2} ds \geq (p-1)(|a| + |b|)^{p-2}|a-b|^2,\end{aligned}$$

since $|c_s| \leq |a| + |b|$ and $p - 2 \leq 0$. For the Hölder bound let $\delta = |a - b|$ and assume $|a| \geq |b|$. If $|a| \leq 2\delta$, then $|T(a) - T(b)| \leq |a|^{p-1} + |b|^{p-1} \leq 2(2\delta)^{p-1} \leq 2^p \delta^{p-1}$. If $|a| > 2\delta$, then $|c_s| \geq |a| - \delta > \delta$ on the whole segment, the operator norm of $DT(c_s)$ is $|c_s|^{p-2} \leq \delta^{p-2}$, and $|T(a) - T(b)| \leq \delta \cdot \delta^{p-2} = \delta^{p-1}$. □

**Theorem B.** *Assume (S) and that $G(x,\cdot)\in C^1([0,\infty))$ with $g(x,0)=0$ for a.e. $x$. Let $(\bar u,\bar\nu,\bar\eta)$ be the limit in Theorem A.*

(a) (Superquadratic type.) *Suppose that for a.e. $x$ the function $t\mapsto k(x,t)=g(x,t)/t$ is nondecreasing on $(0,\infty)$, and that $k(x,t)\ge k_0(t)>0$ for all $t>0$ and a.e. $x$, for some function $k_0$. Then $\nu_n\to\bar\nu$ strongly in $L^2(\Omega;\mathbb{R}^d)$ and $u_n\to\bar u$ strongly in $H^1_0(\Omega)$. Moreover $\nabla\bar u=\nu^*$ for every saddle point, so the minimiser of $J$ over $H^1_0(\Omega)$ is unique.*

(b) (Sub-quadratic power.) *Let $G(x,t)=t^p/p$ with $1<p<2$. Then $\nu_n\to\bar\nu$ and $\nabla u_n\to\nabla\bar u$ strongly in $L^s(\Omega;\mathbb{R}^d)$ for every $s<2$, in particular $u_n\to\bar u$ strongly in $W^{1,p}_0(\Omega)$; moreover $\eta_n\to\bar\eta$ strongly in $L^2(\Omega;\mathbb{R}^d)$, and $u_n\to\bar u$ in $L^2(\Omega)$.*

*Proof.* Apply Theorem A with the saddle point $(u^*,\nu^*,\eta^*)$; by (2.10), $\int_\Omega\psi(x;\nu_n,\nu^*)\,dx=m_n\to0$. Put $e_n=\nu_n-\nu^*$.

(a) Let $\delta>0$. By Lemma 2.12(i), the monotonicity of $k(x,\cdot)$ and the hypothesis at the single value $t=\delta/2$, for a.e. $x$ in the set $\{|e_n|>\delta\}$ we have $\psi(x;\nu_n,\nu^*)\ge\frac12k(x,|e_n|/2)|e_n|^2\ge\frac12k(x,\delta/2)|e_n|^2\ge\frac12k_0(\delta/2)|e_n|^2$, and $\psi\ge0$ everywhere, so

$$\|e_n\|^2\le\delta^2|\Omega|+\frac{2}{k_0(\delta/2)}\,m_n.$$

Hence $\limsup_n\|e_n\|^2\le\delta^2|\Omega|$ for every $\delta>0$, i.e. $\nu_n\to\nu^*$ in $L^2$. Since also $\nu_n\rightharpoonup\bar\nu$, $\nu^*=\bar\nu$ for every saddle point, and $\nabla u_n=s_n+\nu_n\to\bar\nu=\nabla\bar u$ in $L^2$. Finally, every minimiser $u'$ of $J$ on $H^1_0$ belongs to a saddle point: if $J(u')=J(u^*)$, the subgradient inequality and (2.1) give $F(\nabla u')=F(\nabla u^*)+(\eta^*,\nabla(u'-u^*))$, hence $F(q)\ge F(\nabla u^*)+(\eta^*,q-\nabla u^*)=F(\nabla u')+(\eta^*,q-\nabla u')$ for all $q$, i.e. $(u',\nabla u',\eta^*)$ satisfies (2.1). Therefore $\nabla u'=\bar\nu$ and $u'=\bar u$.

(b) By Lemma 2.12(ii) and Hölder's inequality with exponents $2/p$ and $2/(2-p)$,

$$\int_\Omega|e_n|^p\,dx\le\Big(\int_\Omega(|\nu_n|+|\nu^*|)^{p-2}|e_n|^2\,dx\Big)^{p/2}\Big(\int_\Omega(|\nu_n|+|\nu^*|)^p\,dx\Big)^{1-p/2}\le\Big(\frac{m_n}{p-1}\Big)^{p/2}C,$$

where the integrand of the first factor is set to 0 where $\nu_n=\nu^*=0$, and $C$ bounds the second factor because $(\nu_n)$ is bounded in $L^2\subset L^p$. Thus $\nu_n\to\nu^*$ in $L^p$; since $\nu_n\rightharpoonup\bar\nu$, $\nu^*=\bar\nu$. As $(\nu_n)$ is bounded in $L^2$, the interpolation inequality $\|e\|_{L^s}\le\|e\|_{L^p}^{\theta}\|e\|_{L^2}^{1-\theta}$ $(p<s<2)$ gives convergence in $L^s$ for all $s<2$; the same holds for $\nabla u_n=s_n+\nu_n$, and Poincaré's inequality in $W^{1,p}_0$ gives $u_n\to\bar u$ in $W^{1,p}_0$. Finally, $\eta_{n+1}=T(\nu_n)$, $\bar\eta=T(\bar\nu)$, and by Lemma 2.12(ii) and Hölder's inequality (note $2(p-1)\le p$)

$$\|\eta_{n+1}-\bar\eta\|^2\le4^p\int_\Omega|e_n|^{2(p-1)}\,dx\le4^p|\Omega|^{1-\frac{2(p-1)}{p}}\|e_n\|_{L^p}^{2(p-1)}\to0.$$ □

Case (a) contains $t^p/p$ for $p\ge2$, $t^p/p+a(x)t^q/q$ with $2\le p\le q$ and measurable (possibly discontinuous) $a\ge0$ (here $k\ge t^{p-2}$), $t^2\log(1+t)$ $(k(t)=2\log(1+t)+t/(1+t))$, $t^3/\log(1+t)$, and $e^t-1-t$ $(k(t)=(e^t-1)/t\ge1)$, without $\Delta_2/\nabla_2$ conditions. Since $\partial_t(g/t)=(g'-g/t)/t$, for $G\in C^2$ the monotonicity of $k$ is equivalent to $tg'(t)/g(t)\ge1$. When $\operatorname{ess\,sup}_x k(x,1)<\infty$ (all examples just listed, with $a\in L^\infty$), the conclusion of (a) also follows from Glowinski's theorem via Lemma 2.13(b), even for $0<\rho<\frac{1+\sqrt5}{2}r$; (a) is new only when $\operatorname{ess\,sup}_x k(x,1)=\infty$ and $G-\frac{\varepsilon}{2}t^2$ is not convex, e.g. $t^p/p+a(x)t^q/q$ with $2<p\le q$ and $a$ unbounded. Case (b) is not covered by that theorem (Proposition 2.14); it gives strong convergence of $\nu_n$ in $L^s$, $s<2$, not in $L^2$, and on convex plane domains it applies with no further assumption, since there (S) holds for every $f\in L^2$ (Section 3.2). We do not claim strong convergence of $\eta_n$ in case (a).

2.6. **Relation with Glowinski's theorem; the restriction $p\ge2$.** Glowinski [39, Ch. VI, §§4–5] considers $\min_{v\in V}F(Bv)+\Gamma(v)$ with $V,H$ Hilbert spaces, $B\in\mathcal{L}(V,H)$ and $F$, $\Gamma$ proper, convex and lower semicontinuous, and assumes (with $\operatorname{dom}(F\circ B)\cap\operatorname{dom}\Gamma\ne\emptyset$)

(i) the augmented Lagrangian has a saddle point;

(ii) $B$ is injective and $B(V)$ is closed in $H$;
(iii) $F(q)/\|q\| \to +\infty$ as $\|q\| \to \infty$;
(iv) $F = F_0 + F_1$ with $F_0, F_1$ proper, convex, lower semicontinuous, $F_0$ Gâteaux differentiable and uniformly convex on bounded sets in the sense of [39, (4.6)]: for every $M$ there is a continuous strictly increasing $\delta_M$, $\delta_M(0) = 0$, with $(F_0'(q) - F_0'(p), q - p) \geq \delta_M(\|q - p\|)$ for $\|p\|, \|q\| \leq M$.

Then [39, Thm. 5.1], for ALG2 with $0 < \rho < \frac{1+\sqrt{5}}{2} r$ and the steps in our order: $u_n \to u$ strongly in $V$, $\nu_n \to Bu$ strongly in $H$, $\eta_{n+1} - \eta_n \to 0$, $(\eta_n)$ is bounded and its weak cluster points are multipliers; by [39, §7] the whole sequence $(\eta_n)$ converges weakly. For linear $\Gamma$, as here ($\Gamma = -\ell$), the range $0 < \rho < 2r$ due to Gabay and Mercier is recorded in [39, Rem. 5.1, §6.3]. In [39, §6.3] the $s$-Laplacian is set in $W_0^{1,s} \times L^s$, where the theorem "cannot be applied" for $s \neq 2$, and is treated after discretisation. On $V = H_0^1$, $H = L^2$ hypotheses (iii)–(iv) hold for a large class of $G$:

**Lemma 2.13** (Glowinski's hypotheses on $H = L^2$)**.** *Let $G$ be a Young integrand and $F(q) = \int_\Omega G(x, |q|)\, dx$ on $H = L^2(\Omega; \mathbb{R}^d)$.*

(a) *If for some $\varepsilon > 0$ the function $t \mapsto G(x,t) - \frac{\varepsilon}{2} t^2$ is convex on $[0, \infty)$ for a.e. $x$, then $F_0 = \frac{\varepsilon}{2}\|\cdot\|^2$, $F_1 = F - F_0$ satisfy* (iii)–(iv) *with $\delta_M(s) = \varepsilon s^2$.*
(b) *Let $G(x, \cdot) \in C^1$, $g(x,0) = 0$, $k(x,t) = g(x,t)/t$ nondecreasing in $t$, $k_*(t) := \operatorname{ess\,inf}_x k(x,t) > 0$ for $t > 0$ and $K := \operatorname{ess\,sup}_x k(x,1) < \infty$. Let $h(x,t) = G(x,t)$ for $t \leq 1$ and $h(x,t) = G(x,1) + \frac{1}{2} k(x,1)(t^2 - 1)$ for $t > 1$. Then $F_0(q) = \int_\Omega h(x, |q|)\, dx$ and $F_1 = F - F_0$ satisfy* (iii)–(iv)*; $F_0'$ is Lipschitz on $H$ and $(F_0'(q) - F_0'(p), q - p) \geq \frac{3}{8}\, k_*\big(\min\{\frac{\|q-p\|}{4|\Omega|^{1/2}}, 1\}\big)\|q - p\|^2$.*

*Proof.* (a) $\varphi = G(x, \cdot) - \frac{\varepsilon}{2} t^2$ is convex with $\varphi_+'(0) = g(x,0) \geq 0$, hence nondecreasing and $\geq 0$; so $\varphi(x, |\xi|)$ is convex in $\xi$ and $F_1 = \int \varphi(x, |q|)$ is convex, proper and lower semicontinuous (Lemma 2.2). $F \geq F_0$ gives (iii). (b) $\partial_t h(x,t) = t\, k(x, \min\{t, 1\}) \leq Kt$, so $F_0$ is finite and $C^1$ on $H$ with the Lipschitz Nemytskii gradient $q \mapsto k(\cdot, \min\{|q|, 1\})q$. For $t > s \geq 1$, $g(x,t) - g(x,s) \geq (t - s)k(x,s) \geq (t - s)k(x,1)$, so $G - h$ is convex and nondecreasing on $[0, \infty)$ and $F_1$ is convex, proper and lower semicontinuous as in (a). Lemma 2.12(i) with $k(x, \min\{t, 1\})$ and the splitting of the proof of Theorem B(a) with $\delta = \|e\|/(2|\Omega|^{1/2})$, $e = q - p$, give the bound; replacing $k_*$ by its continuous minorant $\underline{k}(\tau) = \frac{1}{\tau}\int_0^\tau k_*$ yields an admissible $\delta_M$ (independent of $M$). Finally $G \geq h \geq \frac{1}{2} k_*(1)(t^2 - 1)$ gives (iii). □

Lemma 2.13 covers $t^p/p$ and $p(x)$ with $2 \leq p(x) \leq p_+ < \infty$ (part (b)), which justifies [4, Thm. 4.3], and all our test integrands except $t^p/p$ with $p < 2$: $e^t - 1$, $e^t - 1 - t$, $t^2/2 + a(x)t^4/4$, $t^{3/2}/(3/2) + t^3/3$ ($g' \geq \frac{3}{2}$) and $t^3/\log(1 + t)$ ($g' \geq 2$, checked numerically) by (a), $t^4/4$ and $t^2 \log(1 + t)$ by (b). For these, under (S), Glowinski's theorem gives strong convergence in $H_0^1 \times L^2$. For $1 < p < 2$:

**Proposition 2.14.** *Let $F(q) = \frac{1}{p}\int_\Omega |q|^p\, dx$ on $H = L^2(\Omega; \mathbb{R}^d)$ with $1 < p < 2$. Then $F$ satisfies neither* (iii) *nor* (iv)*: $F(q)/\|q\| \not\to \infty$, and there is no splitting $F = F_0 + F_1$ as in* (iv).

*Proof.* Fix a unit vector $e$, and for $M$ large a measurable $A_M \subset \Omega$ with $|A_M| = M^{-2+\varepsilon}$, $0 < \varepsilon < 2(2 - p)$; let $q_M = M \mathbf{1}_{A_M} e$. Then $\|q_M\| = M^{\varepsilon/2} \to \infty$ and $F(q_M)/\|q_M\| = M^{p-2+\varepsilon/2}/p \to 0$, so (iii) fails. Next take $|A_M| = M^{-2}$, so $\|q_M\| = 1$; then

$$\frac{F(0) + F(q_M)}{2} - F\Big(\frac{q_M}{2}\Big) = \frac{M^{p-2}}{p}\Big(\frac{1}{2} - 2^{-p}\Big) \to 0.$$

Suppose $F = F_0 + F_1$ as in (iv), and let $m = q_M/2$. By [39, (4.6)] with $M = 1$, for $q \in \{0, q_M\}$ (so $\|q - m\| = \frac{1}{2}$ and the segment lies in the unit ball), $F_0(q) - F_0(m) - (F_0'(m), q - m) = \int_0^1 (F_0'(m + s(q - m)) - F_0'(m), q - m)\, ds \geq \int_0^1 \delta_1(s/2)\frac{ds}{s} \geq \frac{1}{2}\delta_1(\frac{1}{4})$. Averaging over $q \in \{0, q_M\}$ gives $\frac{1}{2}(F_0(0) + F_0(q_M)) - F_0(m) \geq \frac{1}{2}\delta_1(\frac{1}{4}) > 0$, while the same expression for $F_1$ (finite, as $F$ is finite on $L^2$) is $\geq 0$; this contradicts the display. □

Thus for $1 < p < 2$ the classical theorem does not apply, although by Theorems A and B(b) the iteration converges under (S); Theorem A also covers $G$ without any quadratic lower convexity (e.g. with affine pieces), where only weak convergence is proved.

2.7. **What (S) means for the $p$-Laplacian.** We use the following abstract multiplier lemma, also used for Theorem C.

**Lemma 2.15** (Multiplier lemma)**.** *Let $X$ be a real Hilbert space, $R \subset X$ a closed subspace, and $\Psi\colon X \to \mathbb{R}$ convex and bounded above on a ball centred at $q_0 \in R$. If $q_0$ minimises $\Psi$ over $R$, then there is $\zeta \in \partial\Psi(q_0)$ with $\zeta \perp R$.*

*Proof.* For $z \in X$ the difference quotient $s \mapsto (\Psi(q_0+sz)-\Psi(q_0))/s$ is nondecreasing on $s>0$ and bounded below by $\Psi(q_0)-\Psi(q_0-z)$, so $\Psi'(q_0;z) = \lim_{s\downarrow 0}(\Psi(q_0+sz)-\Psi(q_0))/s$ is finite; $z \mapsto \Psi'(q_0;z)$ is positively homogeneous and, by convexity, subadditive; and $\Psi'(q_0;z) \le \Psi(q_0+z)-\Psi(q_0)$. Since $q_0$ minimises $\Psi$ on $R$, $\Psi'(q_0;z) \ge 0$ for $z \in R$. By the Hahn–Banach theorem (extension dominated by a sublinear functional) the zero functional on $R$ extends to a linear functional $\zeta$ on $X$ with $\zeta(z) \le \Psi'(q_0;z) \le \Psi(q_0+z)-\Psi(q_0)$ for all $z$. If $\Psi \le M$ on the ball of radius $\delta$ about $q_0$, then $|\zeta(z)| \le M - \Psi(q_0)$ for $\|z\| \le \delta$, so $\zeta$ is continuous; identifying it with an element of $X$, $\zeta \perp R$ and $\Psi(q_0+z) \ge \Psi(q_0) + (\zeta, z)$, i.e. $\zeta \in \partial\Psi(q_0)$. □

**Proposition 2.16.** *Let $G(t) = t^p/p$ and $d = 2$ (more generally $d \ge 2$ and $p \ge 2d/(d+2)$, so that $W_0^{1,p}(\Omega) \hookrightarrow L^2(\Omega)$). Let $u_p \in W_0^{1,p}(\Omega)$ be the unique minimiser of $J$ over $W_0^{1,p}(\Omega)$.*

(i) *If $1 < p < 2$:* (S) *holds $\iff$ $J$ attains its minimum on $H_0^1(\Omega)$ $\iff$ $u_p \in H_0^1(\Omega)$.*
(ii) *If $p > 2$ and $\nabla u_p \in L^{2p-2}(\Omega)$, then* (S) *holds. Conversely, if $\Omega$ is a Lipschitz domain and* (S) *holds, then $\nabla u_p \in L^{2p-2}(\Omega)$.*

*Proof.* (i) If (S) holds, $J$ attains its minimum on $V = H_0^1$ at $u^*$ (Proposition 2.3). If $J$ attains its minimum on $V$ at $u$, apply Lemma 2.15 with $X = H$, $R = \nabla(V)$ (closed, $\nabla$ being an isometry), $\lambda \in R$ the Riesz representative of $\nabla w \mapsto (f, w)$ on $R$, $\Psi(q) = F(q) - (\lambda, q)$ and $q_0 = \nabla u$; $\Psi$ is finite and bounded on balls because $|q|^p \le 1 + |q|^2$. We get $\zeta \in \partial\Psi(\nabla u)$, $\zeta \perp R$; then $\eta = \zeta + \lambda \in \partial F(\nabla u)$ and $(\eta, \nabla w) = (\lambda, \nabla w) = (f, w)$ for $w \in V$, so $(u, \nabla u, \eta)$ satisfies (2.1). Next, $V$ is dense in $W_0^{1,p}$ (it contains $C_c^\infty$) and $J$ is continuous on $W_0^{1,p}$ (here we use $W_0^{1,p} \hookrightarrow L^2$). If $u$ minimises $J$ on $V$, then $J(v) = \lim J(v_k) \ge J(u)$ for $v \in W_0^{1,p}$ and $v_k \in V$, $v_k \to v$ in $W^{1,p}$, so $u = u_p$ by uniqueness, i.e. $u_p \in V$. Conversely, if $u_p \in V$ it minimises $J$ over $V \subset W_0^{1,p}$.

(ii) The Euler–Lagrange equation $(\eta, \nabla w) = (f, w)$, $\eta = |\nabla u_p|^{p-2}\nabla u_p$, holds for $w \in W_0^{1,p}$, in particular for $w \in C_c^\infty$. If $\nabla u_p \in L^{2p-2}$ then $\eta \in L^2$, both sides are continuous on $V$, and by density the equation holds on $V$; moreover $u_p \in W_0^{1,p} \subset V$, $F(\nabla u_p) < \infty$ and $\eta \in \partial F(\nabla u_p)$ by Lemmas 2.4 and 2.5. For the converse, let $(u^*, \nu^*, \eta^*)$ be a saddle point. Then $F(\nabla u^*) < \infty$, so $\nabla u^* \in L^p$; on a Lipschitz domain a function of $H_0^1$ with gradient in $L^p$ has zero trace and belongs to $W^{1,p}(\Omega)$, hence to $W_0^{1,p}(\Omega)$. Since $u^*$ minimises $J$ on $V \supset W_0^{1,p}$, $u^* = u_p$, and $\eta^* = |\nabla u_p|^{p-2}\nabla u_p \in L^2$ by (2.2), i.e. $\nabla u_p \in L^{2p-2}$. □

Thus for $p < 2$ (S) is equivalent to $u_p \in H_0^1$; for $p > 2$ it is a higher integrability requirement on $\nabla u_p$. When (S) fails, the iterates $u_n \in H_0^1$ are still well defined, but $(\eta_n, \nu_n)$ is unbounded (Proposition 2.11); the behaviour of $u_n$ is open (cf. [41, Rem. 23]).

## 3. When does (S) hold? Regularity, its limits, and counterexamples

3.1. **Two sufficient conditions.**

**Lemma 3.1** (Lipschitz solutions)**.** *Assume that $G(x,\cdot) \in C^1([0,\infty))$ with $g(x,0) = 0$ for a.e. $x$, and that $\operatorname{ess\,sup}_x g(x,M) < \infty$ for every $M > 0$. If $u \in H_0^1(\Omega)$ has $\nabla u \in L^\infty(\Omega)$ and satisfies $\int_\Omega T(x,\nabla u)\cdot\nabla\varphi\,dx = \int_\Omega f\varphi\,dx$ for all $\varphi \in C_c^\infty(\Omega)$, then (S) holds with $(u, \nabla u, T(\cdot,\nabla u))$.*

*Proof.* $\eta = T(\cdot, \nabla u)$ is measurable and bounded, hence in $H$; both sides of the equation are continuous on $V$, so by density it holds for all $w \in V$. $F(\nabla u) < \infty$, and $\eta \in \partial F(\nabla u)$ by Lemmas 2.4 and 2.5(iii). □

**Lemma 3.2** (Explicit flux)**.** *Let $u \in H_0^1(\Omega)$ with $F(\nabla u) < \infty$ and $\eta \in H$ with $\eta(x) \in \partial\Phi(x,\cdot)(\nabla u(x))$ a.e. and $(\eta, \nabla w) = (f, w)$ for all $w \in H_0^1(\Omega)$ (for instance $\eta \in W^{1,\infty}(\Omega;\mathbb{R}^d)$ with $-\operatorname{div}\eta = f$ a.e.). Then (S) holds.*

*Proof.* Lemma 2.4 and Proposition 2.3; for $\eta \in W^{1,\infty}$, $(\eta, \nabla w) = -(\operatorname{div}\eta, w)$ for $w \in C_c^\infty$, hence for $w \in H_0^1$ by density. □

Lemma 3.2 applies to all radial test problems of Section 6, including energies not covered by the regularity theory below (Remark 5.2). (S) does not require a bounded gradient (Proposition 2.16).

3.2. **Regularity results that give (S).** The following results, combined with Lemma 3.1 or Proposition 2.16, yield (S). Where we could not check the hypotheses in the original source (Fan) we say so, and the conclusion (S) is then conditional.

**Literature statement 3.3** (Autonomous Orlicz-type operators; Lieberman, Antonini)**.** Let $G = G(t)$, $g = G' \in C^1(0,\infty)$, $g(0) = 0$, with

$$0 < \delta \le \frac{t\,g'(t)}{g(t)} \le g_0 < \infty \qquad (t > 0). \tag{3.1}$$

Lieberman [44] developed the regularity theory for this structure condition (for $t^p/p$ it reads $\delta = g_0 = p-1$; it implies $\Delta_2$ and $\nabla_2$), with $g$ depending on $t$ only and $(x,u)$ entering through a Hölder perturbation of the vector field of size $\Lambda_1(1+g(|\xi|))(|x-y|^\alpha + |z-w|^\alpha)$ and a lower-order term bounded by $\Lambda_1(1+g(|\xi|)|\xi|)$. His Theorem 1.7 gives *interior* $C^{1,\beta}$ regularity of $W^{1,G}$ solutions with $|u| \le M_0$ (for $f \in L^\infty$ this follows from the $L^\infty$ bound); the corresponding boundary estimates for Dirichlet data are only asserted, on p. 320, by reference to [45, Thms. 1–2]. A global statement is [3, Cor. 1.4]: for $-\operatorname{div} A(x,\nabla u) = f$ whose $\xi$-derivative is controlled by $a(|\xi|) = b(|\xi|)/|\xi|$ with $-1 < i_a \le s_a < \infty$ (equivalent to (3.1) for $b = g$), with $|A(x,\xi) - A(y,\xi)| \le \Lambda(1+b(|\xi|))|x-y|^\alpha$ for a *single* function $b$, on a bounded $C^{1,\alpha}$ domain with $f \in L^\sigma$, $\sigma > d$, and $C^{1,\alpha}$ Dirichlet data, one has $u \in C^{1,\beta}(\overline{\Omega})$.

Consequently, by Lemma 3.1, (S) holds for autonomous $G$ satisfying (3.1) on bounded $C^{1,\alpha}$ domains with $f \in L^\sigma$, $\sigma > d$ (e.g. the disk with constant $f$). Among our test functions, (3.1) holds for $t^p/p$, $t^2/2 + t^4/4$ ($tg'/g \in [1,3]$), $t^{3/2}/(3/2) + t^3/3$ ($[\frac12, 2]$), $t^2\log(1+t)$ ($(1,2]$) and $t^3/\log(1+t)$ ($[1,2)$) (asymptotics at 0 and $\infty$; monotonicity of $tg'/g$ checked numerically on $[10^{-8}, 10^8]$).

**Literature statement 3.4** (Convex domains; Cianchi–Maz'ya)**.** Consider autonomous operators $-\operatorname{div}(a(|\nabla u|)\nabla u) = f$ with $-1 < i_a \le s_a < \infty$, where $i_a = \inf ta'(t)/a(t)$ and $s_a = \sup ta'(t)/a(t)$.

(a) For $d \ge 3$, on convex domains or domains with boundary in $W^2L^{d-1,1}$, and $f \in L^{d,1}$, the gradient is globally bounded [19] (as recalled in [21, Rems. 2.6–2.7]). The version for $p$-type $a$ with $2 \le p < d$, $d \ge 3$, is [20, Thm. 3.5], which also shows that boundedness of the gradient may fail without these domain assumptions.

(b) If $a \in C^1(0,\infty)$, then on bounded convex domains in $\mathbb{R}^d$, $d \ge 2$, with $f \in L^2$, the (unique) generalized solution satisfies $a(|\nabla u|)\nabla u \in W^{1,2}(\Omega)$, with two-sided bounds in terms of $\|f\|$ [21, Thm. 2.3].

Statement (b) gives (S) on convex plane domains for every autonomous $G \in C^2(0,\infty)$ of this class, in particular for $t^p/p$ with every $p > 1$. Indeed, for $d = 2$ and $i_a > -1$, $W_0^{1,G} \subset W_0^{1,2+i_a} \hookrightarrow L^2$, so $f \in L^2$ lies in the dual and the generalized solution is the weak solution [21, Rem. 2.8]; then $\eta = a(|\nabla u|)\nabla u \in W^{1,2}(\Omega) \subset L^s(\Omega)$ for all $s < \infty$ when $d = 2$; since $b(t) = ta(t) \ge a(1)t^{1+i_a}$ for $t \ge 1$, also $\nabla u \in L^s$ for all $s < \infty$, so $u \in H_0^1$, and the Euler–Lagrange equation extends to $H_0^1$ by

density; Lemma 2.4 gives $\eta \in \partial F(\nabla u)$. For $1 < p < 2$ this makes Theorem B(b) unconditional on convex plane domains (Proposition 2.16(i)).

**Literature statement 3.5** (Variable exponent)**.** For $G = t^{p(x)}/p(x)$ with $p \in C^{0,\mu}(\overline{\Omega})$, $\inf p > 1$, $\partial\Omega \in C^{1,\beta}$, bounded weak solutions of the Dirichlet problem are in $C^{1,\alpha}(\overline{\Omega})$, according to Fan [33] as usually cited (hypotheses not read in the original). Interior $C^{1,\beta}$ regularity for Hölder exponents is due to [24, 1]; log-Hölder continuity of $p$ gives Hölder continuity of $u$ and no Lavrentiev gap (cf. [53]), but not, by itself, a bounded gradient.

Under Fan's hypotheses (S) holds by Lemma 3.1 ($g(x, M) \le \max\{1, M\}^{\sup p-1}$).

**Literature statement 3.6** (Double phase: interior results only)**.** Consider the double phase energy $\int(|\nabla w|^p + a(x)|\nabla w|^q)$, $1 < p \le q$, $0 \le a \in C^{0,\alpha}$. For $f \equiv 0$, local minimisers have locally Hölder continuous gradient if $q/p < 1 + \alpha/d$ [22, Thm. 1.2], or $q/p \le 1 + \alpha/d$ [9, Thm. 1.1]; for *bounded* minimisers the condition is $q \le p + \alpha$ [23, 9]; and if $u \in C^{0,\gamma}$, $q < p + \alpha/(1-\gamma)$ suffices [9, Thm. 1.3]. For $a \in W^{1,\sigma}$, $\sigma > d$, De Filippis and Mingione obtain local Lipschitz bounds when $q/p \le 1 + 1/d - 1/\sigma$ (and, if $d = 2$, also $q/p < p$), for $f \in L(d, 1)$ if $d > 2$ and $f \in L^2(\log L)^a$, $a > 2$, if $d = 2$ [25, Thm. 1.3], and Hästö and Ok [42] prove $C^{1,\alpha}_{\mathrm{loc}}$ for minimisers without right-hand side of general generalized Orlicz energies under (VA1), a power-type modulus $\omega(r) \le cr^\beta$ and growth conditions that imply $\Delta_2$. All these results are *interior*.

We know of no refereed global gradient bound for $x$-dependent double phase problems; there (S) must be *assumed* ($u \in H^1_0$ and $|\nabla u|^{p-1} + a|\nabla u|^{q-1} \in L^2(\Omega)$). The indicator coefficient (not Hölder) is not covered at all; the only experiment with a discontinuous coefficient is the radial one of Table 2, where (S) is verified directly.
Energies not covered. For $G = e^t - 1 - t$, $tg'(t)/g(t) = te^t/(e^t - 1) \to \infty$, so (3.1) and $s_a < \infty$ fail ($G \notin \Delta_2$). For $G = e^t - 1$ moreover $g(0) = 1 \ne 0$ and $tg'/g = t \to 0$ at 0. The interior Lipschitz bounds for exponential integrands in [25, Thms. 1.4–1.5] concern $\exp(c_2(x)t^p)$ and iterated exponentials with exponents $p > 1$, so they do not cover these functions as stated. Hence, for general data, *no result known to us gives (S) for* $e^t - 1 - t$ *or* $e^t - 1$; for our radial problems (S) is verified directly (Remark 5.2).

3.3. **Failure of regularity: counterexamples.** The sufficient condition of Lemma 3.1 (a bounded gradient) can fail for non-uniformly elliptic or non-autonomous energies. We have not analysed whether (S) itself fails in the following examples; when there is a Lavrentiev gap (Section 5), the conforming discrete problem is affected regardless of (S). For integrands $C^{0,\alpha}$ in $x$, minimisers have higher integrability ($Du \in L^q_{\mathrm{loc}}$) when $q/p < (d + \alpha)/d$, and this is sharp: for $p < d < d + \alpha < q$ there are $a \in C^{0,\alpha}$ and boundary data with a Lavrentiev gap and a minimiser not in $W^{1,q}_{\mathrm{loc}}$ [32], and local minimisers with singular sets of Hausdorff dimension $> d - p - \varepsilon$ [34]. Zhikov's checkerboard examples [52, 53] exhibit a gap for a variable exponent $< 2$ in two opposite quadrants and $> 2$ in the others, and for double phase with Lipschitz $a$ and $p < 2 < 3 < q$. In $(-1, 1)^d$, $d \ge 2$, Balci, Diening and Surnachev give, with a suitable linear term, a gap for some $a \in C^{0,\alpha}$ when $q > p + \alpha \max\{1, (p-1)/(d-1)\}$ [6, Thm. 32], and for variable exponents with modulus $1/\log(e + 1/t)^\kappa$, $\kappa < 1$ [6, Thm. 30] (arXiv numbering). Unbounded minimisers for autonomous *anisotropic* $(p, q)$ integrands [38, 43], [48, §6] do not concern isotropic $G$ satisfying (3.1); for isotropic $G$ violating it, such as $e^t - 1 - t$, we know of no counterexample and no positive result. Ball and Mizel [8] give one-dimensional strictly convex superlinear ($u$-dependent) integrands whose singular minimiser violates the integrated Euler–Lagrange equation, with a gap.

TABLE 1. When is (S) known to hold? "Via L. 3.1": the cited result gives a bounded gradient up to the boundary. "Conditionally": some hypotheses of the source could not be checked.

| Setting | (S) known? | Reference and hypotheses |
|---|---|---|
| Autonomous $G$, (3.1), $C^{1,\alpha}$ domain, $f \in L^\sigma$, $\sigma > d$ | yes (via L. 3.1) | [3, Cor. 1.4]; [44, Thm. 1.7] is interior, boundary via [45]. $x$-dependence only as Hölder perturbation $\Lambda(1+b(\lvert\xi\rvert))\lvert x-y\rvert^\alpha$ of a single $b$. |
| Autonomous $G$, $\Omega$ convex, $f \in L^{d,1}$, $d \geq 3$ | yes (via L. 3.1) | [19], [20, Thm. 3.5] ($p$-type, $2 \leq p < d$). |
| Autonomous $G \in C^2(0,\infty)$, $\Omega \subset \mathbb{R}^2$ convex, $f \in L^2$ | yes | [21, Thm. 2.3]: $a(\lvert\nabla u\rvert)\nabla u \in W^{1,2}$. |
| $p(x)$, $p \in C^{0,\mu}(\overline{\Omega})$, $\partial\Omega \in C^{1,\beta}$, bounded $u$ | yes (via L. 3.1), conditionally | [33] (hypotheses not read in the original). |
| $p(x)$ log-Hölder only | unknown | Hölder continuity of $u$ only [1]. |
| $p(x)$ continuous, not log-Hölder | no bounded gradient in general | Lavrentiev gap possible [6, Thm. 30]. |
| Double phase, $a \in C^{0,\alpha}$, $q/p \leq 1+\alpha/d$; or bounded $u$, $q \leq p+\alpha$ | interior only; must be assumed | [22, 23, 9] ($f \equiv 0$); [25] ($a \in W^{1,\sigma}$, $\sigma > d$); [42] ($f \equiv 0$). |
| Two-phase, $a$ = indicator function | not covered | $a$ not Hölder; no result applies. |
| Double phase, $q > p+\alpha\max\{1, \frac{p-1}{d-1}\}$, or $p < d < d+\alpha < q$ | no bounded gradient in general; gap | [6, 32, 34, 53]. |
| $e^t-1-t$, $e^t-1$ (no $\Delta_2$) | not available | outside (3.1); not covered by [25, Thms. 1.4–1.5] (exponent $p > 1$ needed). |
| Explicit solutions (radial torsion, manufactured) | yes | Lemma 3.2, by direct verification. |

## 4. The discrete problem and convergence of the discrete iteration

4.1. **Discretisation.** Let $\Omega_h \subset \mathbb{R}^2$ be a polygon (for the disk, the inscribed polygon whose vertices are the boundary nodes) with a conforming triangulation $\mathcal{T}_h$, $x_K$ the barycentre of $K \in \mathcal{T}_h$, and

$$V_h = \{v \in C(\overline{\Omega_h}) : v|_K \in \mathbb{P}_1 \ \forall K,\ v = 0 \text{ on } \partial\Omega_h\}, \qquad H_h = \{q\colon \Omega_h \to \mathbb{R}^2 :\ q|_K \text{ constant } \forall K\},$$

with the $L^2(\Omega_h)$ inner product $(q,q')_h = \sum_K |K|\, q_K \cdot q'_K$ on $H_h$, the norm $\|v\|_{V_h} = \|\nabla v\|_h$ on $V_h$, and $B_h = \nabla\colon V_h \to H_h$. We assume that $G(x,\cdot)$ is a convex Young function for *every* $x \in \overline{\Omega}_h$ (as in all our examples) and that $f$ has well-defined values at the barycentres (e.g. $f \in C(\overline{\Omega_h})$, or piecewise continuous with chosen values at its discontinuities); otherwise one fixes a representative, or defines $\ell_h$ by exact integration. $G$ is sampled at the barycentres and the load is integrated by the barycentre rule:

$$F_h(q) = \sum_K |K|\, G(x_K, |q_K|), \qquad \ell_h(w) = \sum_K |K|\, f(x_K)\, w(x_K), \qquad J_h(v) = F_h(\nabla v) - \ell_h(v).$$

(Any other linear functional $\ell_h$ on $V_h$ may be used; the code also offers a three-point rule.) $\|\cdot\|_{V_h}$ is a norm: if $\nabla v = 0$ then $v$ is constant on each connected component of $\Omega_h$ and vanishes on its boundary. Thus $(V_h, H_h, B_h, F_h, \ell_h)$ satisfies (A1)–(A3), with $F_h$ finite, convex and continuous.

For these data the algorithm (2.3)–(2.5) is what the code computes: step (2.3) is the linear system $rAU_n = b_h + \big[(r\nu_{n-1} - \eta_n, \nabla\phi_i)_h\big]_i$, with $A$ the P1 stiffness matrix on the free nodes (factorised once) and $b_h$ representing $\ell_h$; step (2.4) decouples into one problem per triangle, solved by Lemma 2.7

through a bisection on (2.8) in $[0, |w_K|]$ (in the variable $\log t$, so that very small roots are resolved to relative accuracy); step (2.5) is explicit, and $\eta_{n+1}|_K \in \partial\Phi(x_K, \cdot)(\nu_n|_K)$ is the discrete flux.

4.2. **Theorem C.**

**Theorem C.** *Let $G$ be as in Section 4.1 ($G(x, \cdot)$ a convex Young function for every $x$) and such that $G(x_K, \cdot)$ is superlinear for every $K \in \mathcal{T}_h$:*

$$\lim_{t\to\infty} G(x_K, t)/t = +\infty. \tag{4.1}$$

*Then:*

(i) *$J_h$ has a minimiser $u_h \in V_h$; it is unique if each $G(x_K, \cdot)$ is strictly convex.*

(ii) *There is $\eta_h \in H_h$ such that $(u_h, \nabla u_h, \eta_h)$ is a saddle point of the discrete augmented Lagrangian, for every $r > 0$; that is, $\eta_h|_K \in \partial\Phi(x_K, \cdot)(\nabla u_h|_K)$ for all $K$ and $(\eta_h, \nabla w)_h = \ell_h(w)$ for all $w \in V_h$.*

(iii) *For every $r > 0$, every $\nu_0 \in H_h$ and every $\eta_1 \in \partial F_h(\nu_0)$ (e.g. $\nu_0 = \eta_1 = 0$), the iterates converge: $u_n \to \bar{u}_h$, $\nu_n \to \nabla\bar{u}_h$, $\eta_n \to \bar{\eta}_h$, where $(\bar{u}_h, \nabla\bar{u}_h, \bar{\eta}_h)$ is a discrete saddle point; $\bar{u}_h$ minimises $J_h$, and $\nabla u_n - \nu_n \to 0$. The identity (2.9) holds with $(\cdot,\cdot)_h$, and $E_n$ is nonincreasing.*

This applies to all integrands of this paper, with no regularity assumption and no restriction on $r$. Superlinearity cannot be dropped: for $G \equiv 0$ and $\ell_h \neq 0$, or for $G(t) = t$ and $\ell_h$ large, $J_h$ is unbounded below and no saddle point exists.

*Proof.* (i) Let $L = \sup\{|\ell_h(v)| : \|\nabla v\|_h \le 1\} < \infty$ and $a_{\min} = \min_K |K|$. For $q \in H_h$, $\|q\|_h^2 = \sum_K |K||q_K|^2 \le \max_K |q_K| \sum_K |K||q_K| \le a_{\min}^{-1}\big(\sum_K |K||q_K|\big)^2$, i.e. $\|q\|_{L^1} \ge a_{\min}^{1/2}\|q\|_h$. Given $M > 0$, by (4.1) there is $t_M$ with $G(x_K, t) \ge Mt$ for $t \ge t_M$ and all (finitely many) $K$; since $G \ge 0$, $G(x_K, t) \ge Mt - Mt_M$ for all $t \ge 0$. Hence

$$J_h(v) \ge M\|\nabla v\|_{L^1} - Mt_M|\Omega_h| - L\|\nabla v\|_h \ge \big(Ma_{\min}^{1/2} - L\big)\|\nabla v\|_h - Mt_M|\Omega_h|,$$

and choosing $Ma_{\min}^{1/2} > L$ shows that $J_h$ is coercive on the finite-dimensional space $V_h$. $J_h$ is continuous (each $G(x_K, \cdot)$ is a finite convex function, hence continuous), so it attains its minimum on a nonempty compact sublevel set. If each $\gamma = G(x_K, \cdot)$ is strictly convex, then $\phi = \Phi(x_K, \cdot)$ is strictly convex: $\gamma$ is strictly increasing (if $\gamma(a) = \gamma(b)$ with $a < b$, strict convexity would give $\gamma((a+b)/2) < \gamma(a)$, against monotonicity); for $\xi_1 \neq \xi_2$ with $|\xi_1| = |\xi_2|$ we have $|(\xi_1 + \xi_2)/2| < |\xi_1|$, hence $\phi((\xi_1 + \xi_2)/2) < \phi(\xi_1) = \frac{1}{2}(\phi(\xi_1) + \phi(\xi_2))$, and for $|\xi_1| \neq |\xi_2|$, $\phi((\xi_1 + \xi_2)/2) \le \gamma(\frac{|\xi_1|+|\xi_2|}{2}) < \frac{1}{2}(\phi(\xi_1) + \phi(\xi_2))$; so $F_h$ and, $\nabla$ being injective, $J_h$ are strictly convex.

(ii) Apply Lemma 2.15 with $X = H_h$, $R = \nabla(V_h)$, $\lambda \in R$ with $(\lambda, \nabla w)_h = \ell_h(w)$ for $w \in V_h$, $\Psi(q) = F_h(q) - (\lambda, q)_h$ and $q_0 = \nabla u_h$. $\Psi$ is finite and convex on the finite-dimensional space $H_h$, hence bounded above on a neighbourhood of $q_0$ (on a simplex containing $q_0$ in its interior it is bounded by its values at the vertices). Since $\Psi(\nabla v) = J_h(v)$ for $v \in V_h$, $q_0$ minimises $\Psi$ on $R$. The lemma gives $\zeta \in \partial\Psi(q_0)$, $\zeta \perp R$; then $\eta_h = \zeta + \lambda \in \partial F_h(\nabla u_h)$ and $(\eta_h, \nabla w)_h = \ell_h(w)$. By Proposition 2.3 (which uses only (A1)–(A3)) $(u_h, \nabla u_h, \eta_h)$ is a saddle point, for every $r$. The pointwise form of $\partial F_h$ is the finite-dimensional analogue of Lemma 2.4 ($F_h$ is a sum of functions of the separate blocks $q_K$, and $|K| > 0$).

(iii) By (ii), Assumption (S) holds for the discrete data, and Theorem A, whose proof uses only (A1)–(A3) (Rellich's theorem is only used for the concrete $L^2$ statement), gives weak convergence of $(u_n, \nu_n, \eta_n)$ to a saddle point, $\nabla u_n - \nu_n \to 0$, and (2.9). In finite dimension weak and strong convergence coincide. By Proposition 2.3, $\bar{u}_h$ minimises $J_h$. □

*Remark* 4.1 (Nonhomogeneous Dirichlet data). *Discrete level.* For $u = u_D$ on $\partial\Omega_h$, let $\tilde{u}_h$ be the P1 function with the prescribed boundary values and zero interior values. Writing $u = \tilde{u}_h + v$, $\nu = \nabla\tilde{u}_h + q$, the iteration of the code in the variables $(u, \nu, \eta)$ is the algorithm (2.3)–(2.5) in the

variables $(v, q, \eta)$ for $\tilde{F}_h(q) = F_h(q + \nabla\tilde{u}_h)$, which is finite, convex, superlinear and nonnegative, with $\tilde{F}_h(0) \neq 0$ in general (hence (A2) only asks $F$ to be proper); the initialisation $\nu_0 = \eta_1 = 0$ is admissible, as is any (Remark 2.9). Theorem C(i)–(ii) follows with the same proof; hence Theorem C covers the manufactured test of Section 6.4.

*Continuous level.* If $u = u_D$ on $\partial\Omega$ with $u_D \in H^1(\Omega)$ and $F(\nabla u_D) < \infty$, write $u = u_D + v$, $v \in H^1_0(\Omega)$, and $\tilde{F}(q) = F(q + \nabla u_D)$, $\tilde{\ell}(v) = (f, v)$. Then $(V, H, \nabla, \tilde{F}, \tilde{\ell})$ satisfies (A1)–(A3), and Proposition 2.3 and Theorems A–B apply in the variables $(v, q, \eta)$, with $\nu = q + \nabla u_D$ (the proofs of Theorem B use only $\partial\tilde{F}(q) = \{T(\cdot, q + \nabla u_D)\}$ and $\nu_n - \nu^* = q_n - q^*$). Lemma 3.2 holds with "$u \in H^1_0$" replaced by "$u - u_D \in H^1_0$".

*Remark* 4.2 (Related results and what is not claimed). Theorem C(ii) is classical finite-dimensional convex duality [39, Ch. VI, §4.2], and (iii) follows from [31, Thm. 8] (Kuhn–Tucker pair, full column rank of $\nabla$; see also [14, §3.2]); Glowinski's finite-dimensional [39, Ch. VI, Thm. 5.2] requires a strictly convex $C^1$ part $F_0$. For discontinuous Galerkin discretisations of the $p(x)$-Laplacian, $p(x) \in [p_1, 2]$, Del Pezzo and Martínez [26] apply Glowinski's theorems in finite dimension (their injectivity hypothesis (H4) fails for their lifted gradient and is used only for the convergence of $u_n$; their (H2) quotes Glowinski without the uniform convexity of $F_0$, which is harmless in finite dimension). Theorem C says nothing about the speed of convergence or its dependence on $h$, $p$ or $r$ (cf. [11] and Table 2), nor about the convergence of $u_h$ as $h \to 0$ (Section 5).

## 5. The Lavrentiev phenomenon and the discrete problem

Let $\mathcal{W}$ denote the finite-energy functions in $W^{1,1}_0(\Omega)$ (with the given boundary data), $m_{\mathcal{W}} = \inf_{\mathcal{W}} J$, and $m_{\mathcal{H}}$ the infimum of $J$ over smooth functions with the same boundary data. A *Lavrentiev gap* is the situation $m_{\mathcal{H}} > m_{\mathcal{W}}$, the variational counterpart of the non-density of smooth functions in the energy space ($H \neq W$ in Zhikov's notation [52, 53]). We also use the infimum $m_{\mathrm{Lip}} \geq m_{\mathcal{W}}$ over Lipschitz functions. Examples of a gap were listed in Section 3.3. Conversely, on bounded Lipschitz domains with $a \in C^{0,\alpha}(\overline{\Omega})$, boundary data in $W^{1,q}$ and $q \leq p + \alpha \max\{1, p/d\}$, the infima over $W^{1,p}$, $W^{1,q}$ and $C^\infty_c$ perturbations coincide, without any boundedness assumption [17, Thm. 2.3]; this range is sharp for $p \leq d$ by [6]. Local versions under boundedness or Hölder continuity of $u$ are in [9, Thm. 1.4]; see also the sharp scale of weights in [13]. For variable exponents, log-Hölder continuity excludes the gap (attributed to Zhikov, cf. [53]).

*Consequence for conforming discretisations.* Conforming P1 functions are Lipschitz. For integrands continuous on $\overline{\Omega} \times \mathbb{R}^m \times \mathbb{R}^{m\times d}$, homogeneous Dirichlet data, exact energies on exactly triangulated polyhedral domains and shape-regular meshes, Carstensen and Ortner [18, Thm. 2.1] show that the conforming discrete minima converge to the infimum over Lipschitz functions, so that conforming finite element methods fail exactly when the Lavrentiev phenomenon is present. Balci, Ortner and Storn [7] recall that with exact quadrature conforming minima converge to the infimum over $\mathcal{W} \cap W^{1,\infty}$, observe the resulting energy gap in two-dimensional examples, show that with a "minimum" quadrature even the conforming scheme can converge to $m_{\mathcal{W}}$ [7, §4.2] (so in the gap regime the outcome of our barycentre sampling may depend on the quadrature), and prove, for a specialised one-point quadrature and polynomial growth $1 < p_- < p_+ < \infty$, that Crouzeix–Raviart elements converge to the true minimum [7, Thm. 1]; see also [50, 5]. Hence, in the presence of a gap, *the discrete problem solved by our method need not approximate the continuous minimiser*; Theorem C is not affected, only the approximation property of $u_h$.

*Our test problems.* The following elementary lemma gives a lower bound for every conforming function; it does *not* by itself give convergence of $u_h$ to $u$.

**Lemma 5.1.** *Let $f \in L^\infty(\Omega)$, $u \in W^{1,\infty}(\Omega)$ with $u - u_D \in W^{1,1}_0(\Omega)$, and $\eta \in W^{1,\infty}(\Omega; \mathbb{R}^d)$ with $-\operatorname{div}\eta = f$ and $\eta(x) \in \partial\Phi(x, \cdot)(\nabla u(x))$ a.e. Then $J(v) \geq J(u)$ for every $v \in u_D + W^{1,1}_0(\Omega)$ with $J(v)$ defined; in particular $u$ is a Lipschitz minimiser over the whole finite-energy class, and no function, conforming finite element functions included, has smaller energy.*

*Proof.* The pointwise subgradient inequality gives $\Phi(x,\nabla v)\geq\Phi(x,\nabla u)+\eta\cdot\nabla(v-u)$, and $\int\eta\cdot\nabla w=-\int\operatorname{div}\eta\, w=\int fw$ for $w\in W^{1,1}_0$ (by density of $C^\infty_c$ in $W^{1,1}_0$, both sides being continuous in $W^{1,1}$ since $\eta,\operatorname{div}\eta\in L^\infty$). Integrating with $w=v-u$ gives $J(v)\geq J(u)$. Note that $J(u)$ is finite: the subgradient inequality at $\xi=0$ gives $0\leq\Phi(x,\nabla u)\leq\eta\cdot\nabla u\in L^\infty$. □

Thus $m_{\mathcal{W}}=m_{\mathrm{Lip}}=J(u)$ (no gap between $W^{1,1}$ and Lipschitz functions), and $J(v_h)\geq J(u)$ for every conforming finite element function $v_h$.

*Remark* 5.2 (The radial and manufactured test problems). Consider the unit disk, $f=c>0$, and a radial Young integrand $G(x,t)=G_0(|x|,t)$ such that, as in all our examples, $g_0(s,\cdot)=\partial_tG_0(s,\cdot)$ is continuous and strictly increasing on $(0,\infty)$ and $\inf_s g_0(s,t)\to\infty$ as $t\to\infty$ . Let $\tau(s)=0$ if $cs/2\leq g_0(s,0)$ and otherwise let $\tau(s)>0$ solve $g_0(s,\tau(s))=cs/2$; $\tau$ is measurable, being a monotone inverse ($\tau(s)=\sup\{t\in\mathbb{Q}_{\geq0}:g_0(s,t)<cs/2\}$), and bounded, since $\tau(s)\leq t^*$ whenever $\inf_s g_0(s,t^*)\geq c/2$. The solution is $u(x)=\int_{|x|}^1\tau(s)\,ds$ and the flux is $\eta(x)=-cx/2$. Indeed $|\eta|=c|x|/2$, $\eta$ points in the direction of $\nabla u=-\tau(|x|)x/|x|$, so $\eta(x)\in\partial\Phi(x,\cdot)(\nabla u(x))$ by Lemma 2.5 (part (i) in the dead core of $e^t-1$, where $\nabla u=0$ and $|\eta|\leq1=g(0)$; part (ii) elsewhere, the interface $|x|=1/2$ of the two-phase case being a null set), and $-\operatorname{div}\eta=c$. Since $\tau$ is bounded, $u$ is Lipschitz. By Lemmas 3.2 and 5.1, *for these particular data* (S) holds and there is no Lavrentiev obstruction, also for $e^t-1-t$, $e^t-1$ and $a=\mathbf{1}_{\{|x|>1/2\}}$, which are outside Section 3. Hence Theorem A applies to the *continuous* iteration for every radial test problem, and Glowinski's theorem or Theorem B as indicated in Table 2; the discrete iteration is covered by Theorem C. The same applies to the manufactured solution of Section 6.4 (smooth $u$ and flux; Remark 4.1). These are statements about explicit data only; the sector experiment (Section 6.7) is qualitative.

For the autonomous radial problems, Lemma 5.1 combined with interpolation gives convergence of the exact discrete minimisers.

**Proposition 5.3.** *Let $G=G(t)$ be independent of $x$ and strictly convex, with $g$ continuous and strictly increasing on $(0,\infty)$ and $g(t)\to\infty$; let $\Omega=B_1\subset\mathbb{R}^2$, $f=c>0$, and let $u,\eta$ be as in Remark 5.2. Let $(\mathcal{T}_h)$ be a shape-regular family of triangulations of polygons $\Omega_h\subset B_1$ whose boundary vertices lie on $\partial B_1$, with maximal element diameter $h\to0$, and let $u_h$ be the minimiser of $J_h$ (Theorem C), extended by 0 to $B_1$. Then:*

(i) *$J(u_h)\to J(u)$, and $D_h:=\int_{B_1}\big[\Phi(\nabla u_h)-\Phi(\nabla u)-\eta\cdot\nabla(u_h-u)\big]\,dx=J(u_h)-J(u)\to0$;*
(ii) *$\nabla u_h\to\nabla u$ in measure in $B_1$;*
(iii) *if moreover $g(0)=0$ and $k(t)=g(t)/t$ is nondecreasing and positive on $(0,\infty)$, then $u_h\to u$ in $H^1_0(B_1)$.*

*Proof.* For $v\in V_h$ extended by 0, $J_h(v)=J(v)$: $G$ does not depend on $x$ and $\nabla v$ is piecewise constant, so $F_h(\nabla v)=\int_{B_1}G(|\nabla v|)$ (recall $G(0)=0$), and the barycentre rule is exact for $\int_K cv$ because $v|_K$ is affine. Moreover $v\in W^{1,\infty}_0(B_1)$.

(i) By Lemma 5.1, $J(u_h)\geq J(u)$, and integrating the definition of $D_h$ with $\int\eta\cdot\nabla w=\int cw$ ($w\in W^{1,1}_0$) gives $D_h=J(u_h)-J(u)\geq0$. The nodal interpolant $I_hu$ belongs to $V_h$ because $u=0$ on $\partial B_1$, so $J(u_h)=J_h(u_h)\leq J_h(I_hu)=J(I_hu)$. Let $L=\sup\tau$ be the Lipschitz constant of $u$. On each triangle, $\nabla I_hu$ is determined by the differences of $u$ along two edges, each bounded by $L$ times the edge length, so $|\nabla I_hu|\leq C_\sigma L$ with $C_\sigma$ depending only on the shape regularity (as in [18, Lemma 2.1 and proof of Thm. 2.1]). Since $g$ is continuous and strictly increasing, $\tau$ is continuous on $[0,1]$ and $\nabla u(x)=-\tau(|x|)x/|x|$ is continuous on $\overline{B_1}\setminus\{0\}$; by the mean value theorem along the edges, $|\nabla I_hu|_K-\nabla u(x)|\leq C_\sigma\operatorname{osc}_K\nabla u$ for $x\in K$, which tends to 0 for every $x\neq0$ in $B_1$ (for $h$ small, $x\in\Omega_h$, because the boundary edges have length $\leq h$). Also $|B_1\setminus\Omega_h|\to0$ and $|I_hu|\leq\sup|u|$. By dominated convergence (majorant $G(C_\sigma L)+c\sup|u|$), $J(I_hu)\to J(u)$. Hence $J(u)\leq J(u_h)\leq J(I_hu)\to J(u)$.

(ii) Let $\varepsilon > 0$ and $Q = \{(a,\zeta) : |a| \le L,\ |\zeta| \le c/2,\ \zeta \in \partial\Phi(a)\}$, a compact set (the graph of $\partial\Phi$ is closed, $\Phi$ being continuous and convex), which contains $(\nabla u(x), \eta(x))$ for a.e. $x$. For $(a,\zeta) \in Q$ the convex function $D_{a,\zeta}(b) = \Phi(b) - \Phi(a) - \zeta\cdot(b-a)$ is $\ge 0$ and vanishes at $a$; hence it is nondecreasing along rays from $a$, and $D_{a,\zeta}(b) \ge \mu(\varepsilon) := \min\{D_{a,\zeta}(z) : (a,\zeta)\in Q,\ |z-a| = \varepsilon\}$ when $|b-a| \ge \varepsilon$. The minimum is attained and positive: if $D_{a,\zeta}(z) = 0$ with $z \ne a$, then $\Phi(\frac{a+z}{2}) \ge \Phi(a) + \zeta\cdot\frac{z-a}{2} = \frac12(\Phi(a)+\Phi(z))$, contradicting the strict convexity of $\Phi$ (proved in Theorem C(i)). Therefore $|\{|\nabla u_h - \nabla u| \ge \varepsilon\}| \le D_h/\mu(\varepsilon) \to 0$.

(iii) Now $\Phi \in C^1$ and $\eta = T(\nabla u)$ (Lemma 2.5(iii)). With $a = \nabla u(x)$, $b = \nabla u_h(x)$, and $\psi$ as in Section 2.5,

$$\begin{aligned}\Phi(b) - \Phi(a) - T(a)\cdot(b-a) &= \int_0^1 \big(T(a+s(b-a)) - T(a)\big)\cdot(b-a)\,ds \\ &= \int_0^1 \psi\big(a+s(b-a),a\big)\frac{ds}{s} \ge \frac18\,k\Big(\frac{|b-a|}{4}\Big)|b-a|^2,\end{aligned}$$

by Lemma 2.12(i) and the monotonicity of $k$, restricting the integral to $s \in [\frac12, 1]$. Splitting $B_1$ into $\{|\nabla u_h - \nabla u| \le \delta\}$ and its complement as in the proof of Theorem B(a) gives $\|\nabla(u_h - u)\|^2 \le \delta^2|B_1| + 8D_h/k(\delta/4)$, so $\nabla u_h \to \nabla u$ in $L^2(B_1)$. □

Parts (i)–(ii) apply to all autonomous radial test problems of Table 2, and part (iii) to $t^2/2$, $t^4/4$, $t^2/2 + t^4/4$, $t^2\log(1+t)$, $t^3/\log(1+t)$ and $e^t - 1 - t$. The proposition concerns the exact discrete minimisers, not the computed iterates, and gives no rate. It does not cover the two-phase problem ($J_h \ne J$ on triangles cut by the interface, and we have no uniform bound on $\nabla u_h$).

## 6. Numerical experiments

6.1. **Protocol.** All computations use the package `glaplace` (P1 elements, piecewise constant $\nu, \eta$, $G$ sampled at barycentres, one sparse LU factorisation per problem, vectorised bisection for (2.8)). The disk is meshed by an unstructured DistMesh-type generator with target edge lengths $h = 0.2, 0.1, 0.05, 0.025, 0.0125$ (88 to 23 227 nodes; actual maximal edge lengths 0.288, 0.136, 0.0625, 0.0313, 0.0161); $\Omega_h$ is the inscribed polygon. The iteration starts from $\nu_0 = \eta_1 = 0$ and stops when the scaled primal residual $\|\nabla u_n - \nu_n\|_h/(1 + \|\nabla u_n\|_h)$, which tends to 0 by Theorem C, is below a tolerance, or after a maximum number of iterations. Errors are *true* finite element errors: $\|u - u_h\|_{L^2(\Omega_h)}$ and $\|\nabla(u - u_h)\|_{L^2(\Omega_h)}$ are computed with a six-point degree-4 Dunavant rule on every triangle, relative to the corresponding norm of $u$; the maximum error is taken at the nodes (absolute). Orders $\kappa$ in error $\approx Ch^\kappa$ are least-squares fits in the target $h$ over the three finest meshes. The exact radial solutions $u(x) = \int_{|x|}^1 g^{-1}(cs/2)\,ds$ are computed by bisection and Gauss–Legendre quadrature; they deviate from the closed forms for $t^p/p$, $p = 1.5, 2, 4$, by at most $9\times10^{-9}$ (relative), and were not checked independently for the other $G$. We do not use the nodal difference $I_h u - u_h$, which is superclose: its $L^2$ norm underestimates the true error by factors 7–25 on the finest mesh and suggests spurious orders 2.1–2.5. Scripts and raw data are distributed with the code (see *Software availability*).

6.2. **Torsion on the disk with exact solutions.** We solve (1.2) on the unit disk with $f = c$ and $u = 0$ on the boundary, for the ten Young functions in the upper part of Table 2: $c = 1$, $r = 1$, tolerance $10^{-9}$, at most 3000 iterations, except for $e^t - 1$ ($c = 8$, $r = 30$), whose solution has a dead core $\{|x| \le 1/4\}$ where $\nabla u = 0$. The two-phase case uses $G(x,t) = t^2/2 + a(x)t^4/4$ with $a = \mathbf{1}_{\{|x|>1/2\}}$, an interface not aligned with the mesh. The errors are shown in Figure 1.

*Observations.* (1) For smooth $G$ the observed orders are 2 for $\|u-u_h\|_{L^2}$ and 1 for $\|\nabla(u-u_h)\|_{L^2}$, independently of $G$ (including $e^t - 1 - t$, $t^3/\log(1+t)$ and $t^{1.5}/1.5$); on the finest mesh the relative errors are between $6.0\times10^{-5}$ and $1.2\times10^{-4}$ for $u$ and between $4.9\times10^{-3}$ and $7.5\times10^{-3}$ for $\nabla u$. (2) The nodal maximum error has order about 2 in most cases, but only 1.1–1.4, with fluctuating

TABLE 2. Fitted orders of the relative $L^2$ errors of $u$ and $\nabla u$ and of the nodal maximum error, and number of iterations on the coarsest and on the finest mesh. Top: radial torsion, $h = 0.2, \dots, 0.0125$. Middle: $t^p/p$, $p$ close to 1, $h = 0.2, \dots, 0.025$, at most 6000 iterations (Section 6.3). Bottom: manufactured solution, $n = 10, \dots, 80$, pairwise orders between $n = 40$ and $n = 80$ (Section 6.4). Column "theory": the result that applies to the *continuous* iteration, given that (S) holds for these data (Remark 5.2); the discrete iteration is covered by Theorem C in all cases. "G": Glowinski's Theorem 5.1 applies via Lemma 2.13. "L": $G$ satisfies (3.1). *: tolerance not reached. †: two of the three meshes used in the fit did not reach the tolerance.

| case | $G$ | theory | order | | | iterations |
|---|---|---|---|---|---|---|
| | | | $L^2(u)$ | $L^2(\nabla u)$ | max | coarsest/finest |
| p1.5 | $t^{1.5}/1.5$ | B(b), L | 2.01 | 1.02 | 1.94 | 147/837 |
| p2 | $t^2/2$ | G, B(a), L | 2.00 | 1.02 | 2.02 | 29/29 |
| p4 | $t^4/4$ | G, B(a), L | 2.00 | 1.00 | 1.17 | 47/176 |
| pq24 | $t^2/2 + t^4/4$ | G, B(a), L | 2.00 | 1.01 | 2.02 | 30/26 |
| pq1.5_3 | $t^{1.5}/1.5 + t^3/3$ | G, L | 2.01 | 1.02 | 1.98 | 148/843 |
| plog2_1 | $t^2 \log(1+t)$ | G, B(a), L | 2.00 | 1.01 | 1.37 | 36/60 |
| plog3_-1 | $t^3/\log(1+t)$ | G, B(a), L | 2.00 | 1.02 | 2.02 | 41/33 |
| exp | $e^t - 1 - t$ | G, B(a) | 2.00 | 1.01 | 2.02 | 29/25 |
| twophase | $t^2/2 + \mathbf{1}_{\lvert x\rvert > \frac{1}{2}} t^4/4$ | G, B(a) | 1.90 | 0.98 | 0.79 | 30/27 |
| expdead† | $e^t - 1$, $c = 8$, $r = 30$ | G | 1.97 | 1.01 | 1.11 | 252/3000* |
| p1.2 | $t^{1.2}/1.2$ | B(b), L | 1.98 | 1.03 | – | 6000*/6000* |
| p1.1 | $t^{1.1}/1.1$ | B(b), L | 1.84 | 1.02 | – | 6000*/6000* |
| manufactured | $t^2/2 + (1+x_1)t^4/4$ | G, B(a) | 2.00 | 1.00 | – | 163/218 |

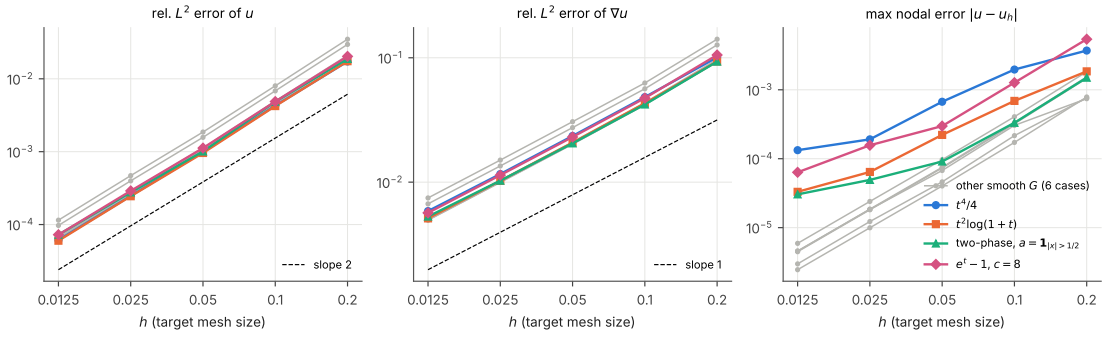



FIGURE 1. Radial torsion: relative $L^2$ errors of $u$ (left) and $\nabla u$ (centre), and nodal maximum error (right), versus the target mesh size. Gray: the six cases $t^{1.5}/1.5$, $t^2/2$, $t^2/2 + t^4/4$, $t^{1.5}/1.5 + t^3/3$, $t^3/\log(1+t)$, $e^t - 1 - t$; the four cases with reduced maximum-norm order are highlighted.

pairwise orders, for $t^4/4$, $t^2\log(1+t)$ and $e^t - 1$; we attribute this, without proof, to the limited smoothness of $u$ at the origin ($|\nabla u| \sim |x|^{1/3}$ for $p = 4$). (3) For the two-phase coefficient, whose jump at $|x| = 1/2$ is not resolved by the mesh and is sampled at barycentres, the orders drop to 1.90 ($L^2$) and 0.79 (maximum), consistently with the jump of $u'$ across the interface. (4) For $e^t - 1$ the iteration is slow ($\partial F$ is multivalued; Figure 2): on the two finest meshes the tolerance was not reached in 3000 iterations (residuals $1.9 \times 10^{-8}$, $3.9 \times 10^{-8}$), but the errors are in line with the

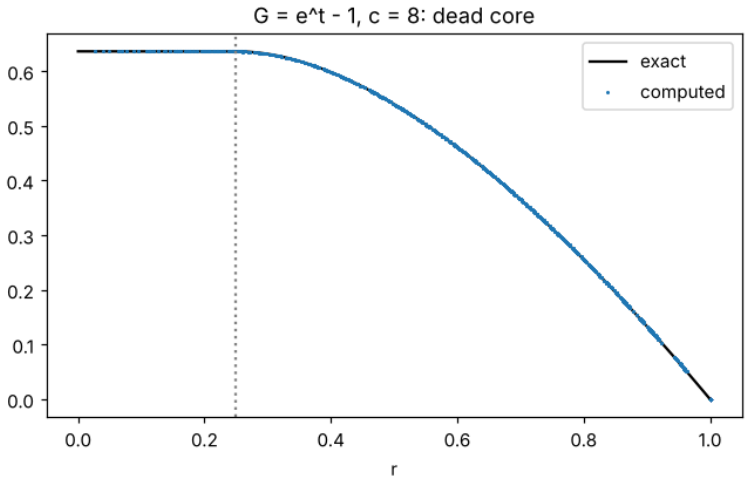



FIGURE 2. $G = e^t - 1$, $c = 8$: exact radial solution (line) and nodal values of $u_h$ ($h = 0.05$, $r = 30$) against $|x|$; the dotted line marks the boundary $|x| = 2/c = 1/4$ of the dead core.

coarser meshes; the computed solution is flat in the dead core ($|\nabla u_h| \le 1.6 \times 10^{-7}$ on triangles with barycentre in $\{|x| < 0.2\}$, $h = 0.05$, all $r$ of Section 6.5). (5) The number of iterations grows as $h$ decreases for $t^{1.5}/1.5$ (roughly like $h^{-1/2}$), $t^{1.5}/1.5 + t^3/3$, $t^4/4$, $t^2 \log(1+t)$ and $e^t - 1$, and is essentially independent of $h$ otherwise.

6.3. **The $p$-Laplacian for $p$ close to 1.** For $t^p/p$ with $p = 1.2$ and $p = 1.1$ ($c = 1$, $r = 1$, tolerance $10^{-9}$, at most 6000 iterations, $h = 0.2, \dots, 0.025$) *none* of the eight runs reached the tolerance; the final residuals were between $1.15 \times 10^{-7}$ and $4.1 \times 10^{-7}$ for $p = 1.2$ and between $6.3 \times 10^{-7}$ and $3.9 \times 10^{-6}$ for $p = 1.1$. On the two finest meshes the $L^2$ error changes between iterations 3000 and 6000 by less than $3 \times 10^{-5}$ (relative) for $p = 1.2$, but by about 3% for $p = 1.1$, whose errors thus contain an iteration component of a few percent; with this caveat, the fitted orders are those of Table 2. The error constants deteriorate as $p \to 1$: on the mesh $h = 0.025$ the relative $L^2$ error is $1.6 \times 10^{-3}$ for $p = 1.2$ and $6.0 \times 10^{-3}$ for $p = 1.1$, against $4.7 \times 10^{-4}$ for $p = 1.5$. We have no explanation for the slow convergence of the iteration as $p \to 1$ beyond the fact that $\partial F$ becomes multivalued in the limit.

6.4. **An $x$-dependent Young function.** On $(0,1)^2$ with uniform meshes of $n \times n$ squares (two triangles each, $h = 1/n$) we take $G(x,t) = t^2/2 + (1 + x_1)t^4/4$, $u = e^{x_1+x_2}$, $f = -\operatorname{div}(g(x, |\nabla u|)\nabla u/|\nabla u|)$ computed symbolically, Dirichlet data $u$, a three-point load rule, $r = 100$ and tolerance $10^{-10}$. Here (S) holds by Lemma 3.2 and Remark 4.1, Theorem B(a) and Glowinski's theorem apply ($k(x,t) = g(x,t)/t = 1 + (1 + x_1)t^2 \ge 1$), and Theorem C applies by Remark 4.1. The pairwise orders are 2.015, 2.004, 2.001 in $L^2$ and 1.001, 1.000, 1.000 for the gradient (relative errors $1.10 \times 10^{-5}$ and $3.61 \times 10^{-3}$ for $n = 80$).

6.5. **The role of the parameter $r$.** By Theorem C the discrete iteration converges for every $r > 0$, but the speed depends strongly on $r$. For the manufactured problem ($n = 40$, tolerance $10^{-10}$) the iteration counts for $r = 1, 10, 30, 100, 300$ are $> 5000$ (not converged), 1167, 396, 208, 516, with relative $L^2$ errors between $4.413 \times 10^{-5}$ and $4.414 \times 10^{-5}$ in all converged runs. For the dead core problem ($e^t - 1$, $c = 8$, $h = 0.05$, tolerance $10^{-8}$) the counts for $r = 1, 3, 10, 30, 100$ are $> 6000$ (not converged), 2139, 650, 244, 529. Heuristically, the best $r$ is of the order of $g(t)/t$ at the typical size $t$ of $|\nabla u|$; we have no theory for this. With $r = 100$ the dead core error at termination

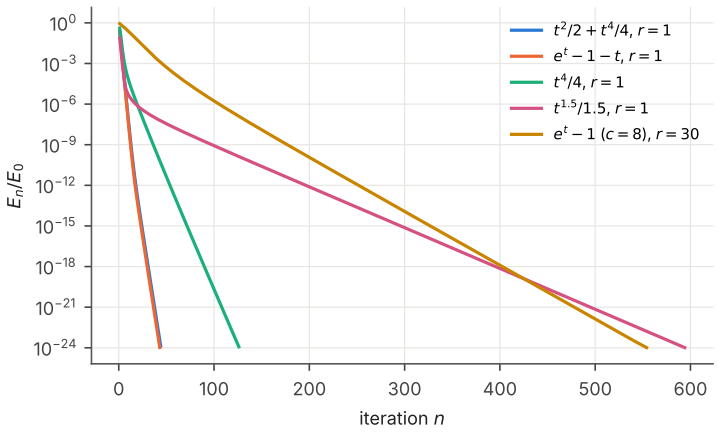



FIGURE 3. Decay of the Lyapunov functional $E_n/E_0$ of Theorem A for the discrete iteration on the disk mesh $h = 0.1$, plotted until $E_n/E_0$ reaches $10^{-24}$.

is $1.1292 \times 10^{-3}$, against $1.1204$–$1.1205 \times 10^{-3}$ for $r \le 30$: for large $r$ a small primal residual does not guarantee a small iteration error. $\|\nu_n - \nu_{n-1}\|$, which appears in (2.9) and bounds the ADMM dual residual $r\nabla^*(\nu_n - \nu_{n-1})$, should also be monitored [14]; we have not rerun the experiments with it.

6.6. **Numerical check of the Lyapunov identity.** On the disk mesh $h = 0.1$ we ran 1500 or 2500 iterations for $t^2/2 + t^4/4$, $t^{1.5}/1.5$, $t^4/4$, $e^t - 1 - t$ ($r = 1$) and $e^t - 1$ ($c = 8$, $r = 30$), with the proxy $u^* = u_L$, $\nu^* = \nabla u_L$, $\eta^* = \eta_{L+1} + r(\nu_L - \nu_{L-1})$, which satisfies all that (2.9) uses up to round-off. The identity then only checks the implementation (residual $\le 7.5 \times 10^{-16} E_0$); the substantive content is the sign of the two terms required to be nonnegative: as long as $E_{n-1}/E_0 > 10^{-20}$, $E_n$ strictly decreases and both terms are positive in all five cases, including the nonsmooth $e^t - 1$. The decay of $E_n$ appears geometric, with rates depending strongly on $G$ and $r$ (Figure 3).

6.7. **A two-phase problem with Hölder coefficient.** We take on the unit disk $f = 4$, $u = 0$ on the boundary and $G(x,t) = t^2/2 + a(x)\,t^3/3$ with $a(x) = \operatorname{dist}(x, S)$, $S = \{x_1 \ge 0,\ x_2 \ge 0\}$: $a$ is Lipschitz and vanishes on the quarter-disk sector $S \cap \Omega$ (quadratic phase there, $p = 2$, $q = 3$, $\alpha = 1$, so that $q \le p + \alpha$). This is the regime covered by the double phase theory (Section 3) and, by [17, Thm. 2.3], there is no Lavrentiev gap ($q \le p + \alpha \max\{1, p/d\}$), so conforming P1 elements are not obstructed. We use $r = 3$, tolerance $10^{-9}$ and $h = 0.2,\ 0.1,\ 0.05,\ 0.025$ (88 to 5809 nodes). No exact solution is available, so we only monitor stability under refinement. The iteration converges in 60 iterations on every mesh. The discrete energy $J_h(u_h)$ is $-2.3758$, $-2.4134$, $-2.4226$, $-2.4247$, with successive differences decreasing by a factor of about 4; $\max u_h$ is 0.7629, 0.7685, 0.7704, 0.7706 and $\max|\nabla u_h|$ is 1.657, 1.757, 1.787, 1.807, i.e. bounded and stabilising, as expected when (S) holds with $\nabla u \in L^\infty$ (Figure 4). This is consistent with, but does not prove, (S) up to the boundary: the interior regularity results of Section 3 do not cover the boundary.

6.8. **Comparison with the literature on rates; limitations.** Proved a priori estimates for P1 elements are of order $h$ in $W^{1,p}$-type norms or quasi-norms, under regularity assumptions on the solution: for the $p$-Laplacian [10, 47, 30]; for N-functions of class $\Delta_2 \cap \nabla_2$, for $\|\mathcal{V}(\nabla u) - \mathcal{V}(\nabla u_h)\|_{L^2}$ with $\mathcal{V}(a) = (\varphi'(|a|)/|a|)^{1/2} a$, when $\mathcal{V}(\nabla u) \in W^{1,2}$ [29, 28]; for $p(x)$, see [27, 15]. Glowinski and Marrocco [40] prove convergence without rate and $W^{1,p}$ rates for $u \in W^{2,p}$; the $L^2$ estimate in [47, Thm. 4.3] follows from the energy estimate by Hölder's inequality. We have found no proof of an $O(h^2)$ estimate in $L^2$ for the $p$- or the $g$-Laplacian (a duality argument is not available in general in the degenerate or singular case), so the observed orders go beyond the proved theory; they are

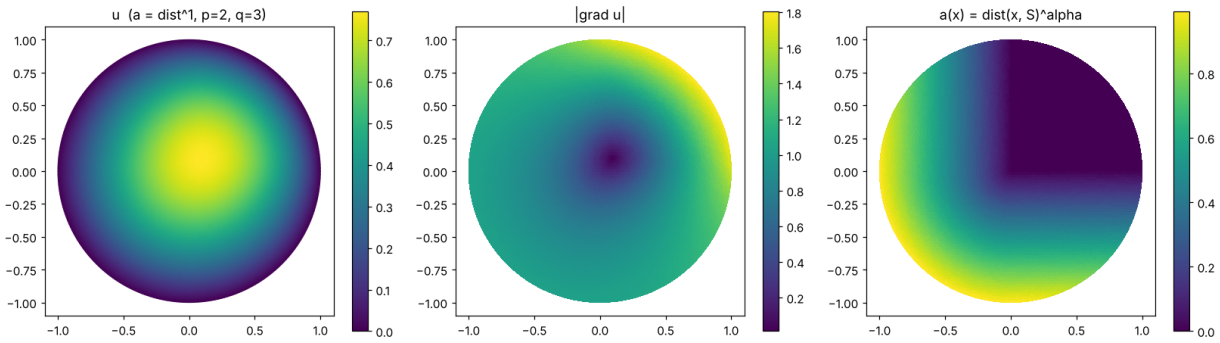



FIGURE 4. Two-phase problem $t^2/2 + a(x)t^3/3$, $a = \operatorname{dist}(x, S)$, on the disk, $f = 4$, $h = 0.025$. Left: $u_h$; centre: $|\nabla u_h|$ (piecewise constant); right: $a$ sampled at barycentres.

consistent with linear heuristics where $g(x,t)/t$ is bounded above and below on the range of $|\nabla u|$. Previous computations with the method report errors on a single mesh [4] or, for discontinuous Galerkin discretisations, $L^2$ errors on a sequence of meshes without fitted orders [26].

Limitations: (i) orders are fitted over three unstructured meshes in the target $h$, and pairwise orders fluctuate; (ii) errors are computed on the inscribed polygons $\Omega_h$; (iii) the stopping test uses the primal residual only (Section 6.5); (iv) twelve runs did not reach the tolerance (two for $e^t - 1$, eight for $p \in \{1.1, 1.2\}$, the $r = 1$ runs of Section 6.5), and the fits for $e^t - 1$ and $p \in \{1.1, 1.2\}$ use them; (v) the "exact" radial solutions are computed numerically; (vi) the sector example has no exact solution, and all others are smooth or explicit, so the experiments say nothing about the Lavrentiev regime.

## 7. Conclusions and open problems

For $\rho = r$ and an arbitrary convex Young integrand, convergence at the continuous level is equivalent to boundedness and hinges on (S), which is not available in general for exponential and double phase energies, whereas the finite element iteration converges unconditionally. Open problems include: (1) the behaviour of $u_n$ when (S) fails (then $(\eta_n, \nu_n)$ is unbounded, Proposition 2.11; cf. [41, Rem. 23]), for instance for $1 < p < 2$ when the minimiser is not in $H^1_0$, or for $p > 2$ when $\nabla u \notin L^{2p-2}$; (2) strong convergence of $u_n$ for general convex $G$ (the counterexample of [16] concerns $\eta_n$ in the abstract setting), the case $\rho \neq r$ without (iii)–(iv), and rates for the iteration, including their dependence on $h$, $p$ and $r$ and a theory for the choice of $r$; (3) convergence of the P1 minimisers $u_h$ to $u$ beyond the autonomous radial case of Proposition 5.3 (e.g. for discontinuous coefficients), and a priori error estimates explaining the observed $O(h^2)$ behaviour in $L^2$ for degenerate or singular $G$ and for $G$ without $\Delta_2$; (4) Lavrentiev-aware discretisations, for instance a Crouzeix–Raviart version of the method in the spirit of [7], for which an analogue of Theorem C would hold with the same proof; (5) global regularity, hence (S), for Musielak–Orlicz $G(x,t)$ without $\Delta_2$ and for double phase energies up to the boundary.

**Software availability.** The Python package `glaplace` (NumPy/SciPy; MIT license), which implements the method for arbitrary Young functions on general two-dimensional triangulations and contains the scripts for all experiments of Section 6, is available at `https://github.com/jfbonder/glaplace`. It generalises the code for the $p(x)$-Laplacian of [4].

**Declaration of the use of AI tools.** Generative AI was used extensively in this work, in the form of Claude models developed by Anthropic (Claude Sonnet and Claude Opus, used through the Claude desktop application and its agent tools, October 2026). The research problem, the choice

of method and of the test problems, and the scientific direction are the authors'. The AI tools were used for: (i) writing the Python code of the solver and of all numerical experiments, including the tests; (ii) drafting the proofs of Theorems A, B and C and of the auxiliary results, starting from the authors' earlier unpublished argument at the continuous level; (iii) searching and compiling the literature and extracting the hypotheses of the cited results from the sources; (iv) drafting and editing the text of the article; and (v) independent critical reviews of the mathematics and of the exposition, carried out by separate instances of the model. The numerical results reported here were produced by running the code described above. The authors have read the article and the code, take full responsibility for their content, and are responsible for checking the proofs and the accuracy of every citation.

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